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The Third Enclosure Movement

Rethinking AI Regulation through a Taxonomy of Restricted Knowledge

PETER KAHL

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About the Publisher

Lex et Ratio Ltd provides research, advisory, and strategic consulting in governance reform, fiduciary accountability, and epistemic ethics, integrating legal analysis, institutional theory, and practical reform strategies across public, corporate, and academic institutions.

Preface

This study originated as my LLM dissertation at the University of Reading, where it was completed under the supervision of **Dr Jorge Guira**, whose guidance and critique were invaluable throughout its development. The work now forms part of the *Lex et Ratio Working Paper Series* and serves as the conceptual foundation for ongoing research into fiduciary-epistemic governance and AI regulation.

— Peter Kahl, November 2025

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Abstract

This dissertation examines how law and policy can respond to the multi-layered enclosures emerging in AI governance. It introduces two conceptual tools: the *Third Enclosure Movement*, situating AI within historical logics of exclusion, and the *Taxonomy of Restricted Knowledge*, distinguishing legal, technical, epistemic, and state-enforced enclosures.

The comparative analysis of the UK, EU, US, and China, alongside case studies in healthcare, agritech, and consumer AI, shows that no single strategy suffices. Abolitionist critiques expose the structural injustices of intellectual property, but treaty lock-in under TRIPS and entrenched domestic frameworks make wholesale abolition infeasible.

By contrast, partial commons reforms can be pursued within current legal architectures. Examples include compulsory licensing under TRIPS (arts 30–31), disclosure and audit obligations under the EU AI Act (arts 16–29), interoperability duties under the Digital Markets Act (arts 6(4)–(9)), and open-licensing conditions for publicly funded AI outputs. These tools embed openness, auditability, and equitable access while preserving incentives for innovation.

The dissertation's contribution is to reframe AI governance as the defence of shared epistemic agency. It argues for a legally realistic middle path: treaty-compliant reforms that constrain recursive enclosure while sustaining democratic legitimacy in the age of AI.

Keywords

AI governance, fiduciary duty, epistemic justice, knowledge regulation, restricted knowledge, intellectual property law, artificial intelligence law, epistemic ethics, information enclosures, legal philosophy, law and technology, governance and regulation

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1. Introduction

1.1 Background and Context

The regulation of artificial intelligence ('AI') is one of the most pressing issues in contemporary legal, political, and ethical discourse. As machine learning systems permeate healthcare, finance, defence, education, and communication, governance is no longer a matter of technical optimisation but of institutional design, political economy, and justice. The debate has shifted from whether to regulate AI to how, and in whose interests.

The central governance challenge concerns control of knowledge—datasets, algorithms, code, and infrastructure. Liberal political economy assumed that temporary exclusive rights incentivised innovation before works entered the public domain. This assumption no longer reflects the realities of AI, which is shaped by layered enclosures: legal rights (IP, trade secrecy), technical architectures (API gating, compute monopolies), epistemic opacity (closed benchmarks, black-box models), and sovereign constraints (data localisation, ideological controls).

Foundational critiques from Rawls, Marx, Ostrom, and Foucault, and contemporary analyses such as Julie Cohen's theory of informational capitalism, demonstrate that questions of knowledge control are inseparable from justice and power.[1] Against this background, I advance the concept of The Third Enclosure Movement, drawing on historical precedents in land and intellectual property to capture the convergence of law, infrastructure, and institutional design in creating entrenched exclusion in the digital domain.

Unlike earlier waves, this enclosure is polycentric and recursive: exclusion is built into technical and legal systems, reinforced by feedback loops between law, design, and incentives. To analyse these dynamics, I employ a Taxonomy of Restricted Knowledge, distinguishing legal, technical, epistemic, and state-enforced forms, which functions as both diagnostic and prescriptive tool.

1.2 Problem Statement

Existing legal frameworks—IP, data protection, competition law, and sectoral rules—were not designed to address the recursive nature of enclosure in AI. Traditional IP regimes rewarded discrete innovations through time-limited exclusivity; AI, by contrast, is iterative, infrastructural, and capital-intensive, dependent on proprietary datasets, closed models, and concentrated compute resources.

[1] John Rawls, *A Theory of Justice: Revised Edition* (Harvard University Press 1999); Karl Marx, *Capital: A Critique of Political Economy, vol I* (Fowkes B tr, Penguin Classics 1990); Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge University Press 1990); Michel Foucault, *Discipline and Punish: The Birth of the Prison* (Sheridan A tr, Vintage Books 1995); Julie E Cohen, *Between Truth and Power: The Legal Constructions of Informational Capitalism* (Oxford University Press 2019).

These dependencies generate durable barriers excluding smaller firms, civil society, and Global South jurisdictions. Enclosure extends beyond IP doctrines into system architecture, opacity, and sovereign regulation. Reform agendas framed around 'AI transparency' or 'responsible innovation' often fail to confront this structure. Even initiatives that begin as open—such as data mandates or interoperability rules—are re-enclosed through throttled APIs, weight redaction, or proprietary benchmarks.

The challenge is therefore to diagnose and respond to overlapping regimes of restricted knowledge with tools that are both institutionally realistic and normatively robust, targeting interventions proportionately to the form and context of enclosure.

1.3 Research Questions and Objectives

Central research question

> Does the rise of AI-driven enclosures necessitate a shift away from purely proprietary regimes toward partial commons-based approaches, and how can the UK, EU, US—and, by contrast, China—reconcile innovation incentives with public knowledge sharing?

Sub-questions

- Access and incentive – How can regulation preserve incentives for capital-intensive AI while lowering barriers for smaller actors, researchers, and civil society?
- Doctrinal adequacy – Are patents, trade secrets, copyright, and database rights sufficient, or are new tools such as data trusts, Commons AI pilots, or audit mandates required?
- Sectoral variation – How do enclosure dynamics differ across healthcare, agritech, and consumer AI, and what lessons follow for regulatory design?
- Comparative divergence – What does China's model of sovereign and ideological enclosure reveal about alternative logics, and how might it challenge liberal jurisdictions?

Objectives

1. Coin and define *The Third Enclosure Movement* as a framework for AI governance.
2. Apply the *Taxonomy of Restricted Knowledge* to comparative and sectoral cases.
3. Evaluate the viability of partial commons reforms as a feasible and defensible alternative to the proprietary paradigm.

1.4 Thesis Statement

The recursive exclusions defining the *Third Enclosure Movement* cannot be resolved by maintaining traditional proprietary IP regimes. Wholesale abolition is neither politically nor economically feasible. This dissertation argues for a partial commons governance model: retaining incentives for capital-intensive AI while conditioning them on transparency, accountability, and equitable access.

This model avoids portraying openness as a universal good or proprietary structures as inherently unjust. Instead, it advances a calibrated strategy, grounded in governance tools tailored to the specific form of enclosure—legal, technical, epistemic, or state-enforced.

1.5 Scope and Jurisdictional Focus

The comparative analysis focuses on four jurisdictions with global regulatory influence and distinct approaches to AI governance. Each illustrates how enclosure is juridically institutionalised:

- **United Kingdom** – The Copyright, Designs and Patents Act 1988 ('CDPA 1988') s 9(3) attributes authorship of computer-generated works to 'the person by whom the arrangements necessary for the creation of the work are undertaken', raising unresolved questions about AI authorship and illustrating legal enclosure through copyright formalism.[2] The Data Protection Act 2018, incorporating the UK GDPR, requires disclosure of 'meaningful information' in certain automated-decision contexts but does not create a free-standing right to explanation.[3]

- **European Union** – The GDPR provides transparency rights but is constrained by trade secrecy defences, illustrating tension between openness and legal enclosure.[4] The Database Directive confers sui generis rights based on investment, enabling exclusion from essential datasets even where no personal data are involved.[5] The Digital Markets Act imposes obligations on 'gatekeepers' to ensure interoperability and access, directly addressing structural and technical enclosure.[6] The Artificial

[2] Copyright, Designs and Patents Act 1988, s 9(3).

[3] Data Protection Act 2018; Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation) [2016] OJ L119/1, as retained in UK law by the European Union (Withdrawal) Act 2018 and amended by Data Protection, Privacy and Electronic Communications (Amendments etc) (EU Exit) Regulations 2019, SI 2019/419.

[4] Regulation (EU) 2016/679, arts 13–15, 22, Recital 71.

[5] Directive 96/9/EC of the European Parliament and of the Council of 11 March 1996 on the legal protection of databases [1996] OJ L77/20.

[6] Regulation (EU) 2022/1925 of the European Parliament and of the Council of 14 September 2022 on contestable and fair markets in the digital sector and amending Directives (EU) 2019/1937 and (EU) 2020/1828 (Digital Markets Act) [2022] OJ L265/1.

Intelligence Act 2024, phased in between January 2025 and July 2027, requires documentation and audit obligations to counter epistemic enclosure.[7]

- **United States** – The Copyright Act 1976, 17 USC, protects software as 'literary works', but the US Copyright Office excludes works lacking human authorship, producing ambiguity resolved by contractual enclosure.[8] The Uniform Trade Secrets Act (UTSA) and Defend Trade Secrets Act 2016, 18 USC § 1836, allow firms to shield datasets, architectures, and evaluation methods from scrutiny, reinforcing epistemic and technical enclosure.[9] Privacy protection remains fragmented: HIPAA, COPPA, FCRA, and GLBA regulate sectoral silos, while the California Consumer Privacy Act 2018 (as amended by the CPRA 2020) grants only limited consumer rights.[10] Competition law tools exist under the Federal Trade Commission Act 1914, 15 USC §§ 41–58, but have only intermittently been applied to digital or algorithmic exclusion.[11]

- **People's Republic of China** – The Cybersecurity Law 2016, Data Security Law 2021, and Personal Information Protection Law 2021 impose localisation and tiered review requirements, enclosing data within sovereign borders.[12] The Provisions on the Management of Algorithmic Recommendations in Internet Information Services 2022 mandate filing and ideological alignment of recommender systems with the Cyberspace Administration of China, exemplifying state-enforced enclosure.[13]

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[7] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 (Artificial Intelligence Act) [2024] OJ L1689/1.

[8] Copyright Act 1976, 17 USC; US Copyright Office, Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence (16 March 2023).

[9] Uniform Trade Secrets Act (US, model law 1979, amended 1985); Defend Trade Secrets Act 2016, 18 USC § 1836.

[10] Health Insurance Portability and Accountability Act 1996; Children's Online Privacy Protection Act 1998; Fair Credit Reporting Act 1970; Gramm–Leach–Bliley Act 1999; California Consumer Privacy Act 2018, Cal Civ Code § 1798.100 et seq.

[11] Federal Trade Commission Act 1914, 15 USC §§ 41–58.

[12] Cybersecurity Law of the People's Republic of China (中华人民共和国网络安全法) (adopted 7 November 2016, effective 1 June 2017) <http://www.npc.gov.cn/zgrdw/npc/xinwen/2016-11/07/content_2001605.htm> accessed 23 August 2025; Data Security Law of the People's Republic of China (中华人民共和国数据安全法) (adopted 10 June 2021, effective 1 September 2021) <http://www.npc.gov.cn/englishnpc/c2759/c23934/202112/t20211209_385109.html> accessed 23 August 2025; Personal Information Protection Law of the People's Republic of China (中华人民共和国个人信息保护法) (adopted 20 August 2021, effective 1 November 2021) <https://www.ssf.gov.cn/portal/zcfg/zcfg/webinfo/2021/11/1640743537111655.htm> accessed 23 August 2025.

[13] Provisions on the Management of Algorithmic Recommendations in Internet Information Services (adopted 31 December 2021, effective 1 March 2022) Cyberspace Administration of China (互联网信息服务算法推荐管理规定) <http://www.cac.gov.cn/2022-01/04/c_1642894606364259.htm> accessed 1 April 2025.

This scope enables comparative mapping of how different systems institutionalise AI enclosure, while also revealing convergences around transparency, competition, and data governance.

1.6 Methodology

This is a socio-legal study: doctrinal analysis is primary, but enriched by comparative method and critical theory. Such integration reflects the LLM programme's emphasis on law in context:

1. Doctrinal analysis of statutory provisions, case law, and regulatory instruments across IP, data protection, competition law, and AI-specific frameworks.

2. Comparative legal method to identify divergences and convergences across the four jurisdictions.

3. Interdisciplinary framing, incorporating Rawlsian egalitarianism, Marxist political economy, Ostromian commons theory, Foucauldian power–knowledge analysis, and Cohen's informational capitalism.

4. Operationalisation of the Taxonomy of Restricted Knowledge (Chapter 2) as both diagnostic and prescriptive.

5. Sectoral case studies in healthcare, agritech, and consumer AI to test the taxonomy's robustness and reform feasibility.

1.7 Significance of the Research

The dissertation makes four principal contributions:

1. Introduces **The Third Enclosure Movement** as an original conceptual framework situating AI governance within the historical lineage of enclosures.

2. Develops the **Taxonomy of Restricted Knowledge** as a practical tool for diagnosing and reforming AI enclosures.

3. Proposes **partial commons-based governance strategies**—including open-licensing carve-outs, fiduciary data trusts, algorithmic audit mandates, and public HPC investment.

4. Offers a **comparative jurisdictional analysis** applying the taxonomy to distinct political–economic configurations.

By uniting doctrinal precision with theoretical depth, this work provides a coherent framework for designing AI governance models that are legally sound, economically viable, and normatively defensible.

2. Taxonomy of Restricted Knowledge

This chapter introduces the *Taxonomy of Restricted Knowledge*, the analytical backbone of the dissertation. It distinguishes four categories of enclosure—legal, technical, epistemic, and state-enforced—and shows how they interact to shape access and control in AI governance. The taxonomy is used throughout the dissertation in comparative analysis (Ch 5), sectoral case studies (Ch 6), and reform proposals (Chs 7–8). As Julie Cohen observes, governance must address the 'layered architectures of informational capitalism', where exclusion operates simultaneously through legal, infrastructural, and discursive means.[14]

2.1 Why a Taxonomy is Needed

Traditional intellectual property analysis identifies barriers to knowledge sharing but does not account for the multi-modal, recursive exclusions of AI governance. Intangible enclosure—control over non-rivalrous assets such as datasets, algorithms, and high-performance computing ('HPC') infrastructure—demands a broader classification capturing how, by whom, and for what purpose access is restricted.

The taxonomy distinguishes four analytically distinct yet overlapping categories:

- **Legal enclosure** – exclusion via formal legal rights (patents, copyright, database rights, trade secrets).
- **Technical enclosure** – exclusion embedded in design or architecture (API gating, infrastructure centralisation).
- **Epistemic enclosure** – exclusion from the means of understanding, verifying, or contesting AI outputs.
- **State-enforced enclosure** – exclusion imposed through sovereign authority (data localisation, export controls, ideological content rules).

Although these categories intersect, separating them analytically enables clearer regulatory diagnosis and response. The taxonomy is applied in three key contexts in this dissertation:

1. Comparative jurisdictional mapping (Chapter 5);
2. Sectoral analysis (Chapter 6);
3. Reform proposals (Chapters 7–8).

[14] Julie E Cohen, *Between Truth and Power: The Legal Constructions of Informational Capitalism* (Oxford University Press 2019) ch 4, 120-125.

2.2 Categories of Restricted Knowledge

The four categories below are defined here for use throughout the dissertation. Later chapters will refer back to these definitions without repeating them.

2.2.1 Legal Enclosure

Legal enclosure arises where statutory, common law, or contractual rights restrict unauthorised access to, reproduction of, or innovation upon AI-related assets. While such regimes incentivise R&D, they also risk 'anticommons' effects where fragmented rights impede collaborative AI innovation.[15] The principal doctrines include:

- **Patents** – At the European Patent Office, AI and machine learning methods are treated as mathematical methods excluded from patentability under art 52(2)(a) EPC, unless the claim pursues a technical purpose or produces a further technical effect.[16] UK law remains aligned through the Patents Act 1977. In the US, the Supreme Court narrowed software patentability in *Alice Corp Pty Ltd v CLS Bank International*,[17] excluding abstract ideas lacking an 'inventive concept'. In China, the Patent Law (as amended 2020, effective 2021) allows AI-related inventions where they solve a technical problem, though abstract mathematical models remain excluded.[18] This doctrinal architecture has facilitated China's rapid rise in AI patent filings.

- **Copyright** – In the UK, computer programs are classified as literary works under the Copyright, Designs and Patents Act 1988, s 1(1)(a). Section 9(3) attributes authorship of computer-generated works to the 'person by whom the arrangements necessary for the creation of the work are undertaken', though this remains untested for autonomous AI. In the EU, Directive 2009/24/EC harmonises protection but requires human authorship. In the US, computer programs are protected under the Copyright Act 1976, 17 USC § 101 et seq, but the US Copyright Office has confirmed that fully machine-generated outputs are ineligible.[19] China's Copyright Law (as amended 2020) follows a similar model, protecting software as literary works without recognising non-human authors.

- **Database rights** – In the UK, sui generis protection arises under the Copyright and Rights in Databases Regulations 1997, SI 1997/3032, which implemented Directive 96/9/EC.[20] The CJEU in *British Horseracing Board Ltd v William Hill Organisation Ltd* confirmed the threshold of 'substantial

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[15] Michael A Heller and Rebecca S Eisenberg, 'Can Patents Deter Innovation? The Anticommons in Biomedical Research' (1998) 280 *Science* 698.

[16] European Patent Office, *Guidelines for Examination in the European Patent Office* (1 March 2025) G-II, 3.3–3.3.1.

[17] 573 US 208 (2014).

[18] Patent Law of the People's Republic of China (adopted 12 March 1984, amended 2020, effective 2021).

[19] US Copyright Office, *Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence* (16 March 2023).

[20] Directive 96/9/EC [1996] OJ L77/20.

investment'.[21] In the EU, the Directive remains the governing instrument. The US provides no sui generis right, relying instead on copyright, contracts, and trade secrecy. China similarly protects original compilations under its Copyright Law, supplemented by the Anti-Unfair Competition Law.

- **Trade secrets** – In the EU, protection is governed by Directive (EU) 2016/943.[22] In the UK, trade secrets remain protected under the common law of confidence[23] and the Trade Secrets (Enforcement, etc.) Regulations 2018, SI 2018/597. In the US, the Uniform Trade Secrets Act (adopted in most states) and the Defend Trade Secrets Act 2016[24] provide parallel protection. In China, trade secrets are covered by the Law Against Unfair Competition (as amended 2019).

Tensions arise where GDPR disclosure duties (arts 13–15, 22) intersect with trade secret protection under Directive 2016/943: controllers often resist transparency by invoking secrecy, leaving courts and regulators to balance disclosure against confidentiality. This unresolved clash exemplifies how legal enclosure can reinforce epistemic opacity.

2.2.2 Technical Enclosure

Technical enclosure occurs when access restrictions are embedded in the infrastructure or architecture of AI systems. In Lessig's terms, 'code is law'—that is, the architecture of cyberspace itself functions as a regulator.[25]

Examples include:

- **API access-gating** – OpenAI's GPT-4 permits access via API but withholds model weights and training datasets, preventing independent replication or audit.

- **Infrastructure centralisation** – Training state-of-the-art models requires substantial HPC resources, often concentrated among a few providers (AWS, Microsoft Azure, Google Cloud), creating capital and partnership bottlenecks.

- **Embedded controls** – Digital rights management ('DRM'), rate-limiting, or interface design that restricts model interrogation or alternative deployments.

Technical enclosure is particularly resistant to legal reform because it is enforced through the system's own operational design. It often overlaps with legal enclosure (e.g. trade secrets reinforcing API restrictions) and epistemic enclosure (e.g. proprietary benchmarks).

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[21] Case C-203/02 [2004] ECR I-10415.

[22] Directive (EU) 2016/943 [2016] OJ L157/1.

[23] *Saltman Engineering Co Ltd v Campbell Engineering Co Ltd* [1948] 65 RPC 203 (CA).

[24] 18 USC § 1836.

[25] Lawrence Lessig, *Code: And Other Laws of Cyberspace* (Basic Books 1999) ch 1, 6–7.

2.2.3 Epistemic Enclosure

Epistemic enclosure arises when the capacity to understand, verify, or contest an AI system is withheld. It concerns not access to outputs but whether outputs can be explained or audited.

Mechanisms include:

- Opaque architectures resistant to interpretability;
- Proprietary benchmarks excluding independent testers;
- Non-replicable performance claims.

In high-stakes contexts—credit, employment, healthcare—this impedes accountability. O'Neil captures the point: 'models are opinions embedded in code'.[26]

Under the GDPR, Articles 13–15 and 22, read with Recital 71, require disclosure of 'meaningful information about the logic involved' in some automated decision-making. The CJEU has interpreted disclosure obligations broadly: in *Nowak v Data Protection Commissioner* (C-434/16) the Court held that even exam scripts could qualify as personal data, underscoring the potentially wide scope of access rights. Yet the European Data Protection Board's guidance confirms that there is no free-standing 'right to explanation'; transparency remains ancillary to existing information rights.[27] In practice, the force of these provisions is blunted by vague statutory terms, under-resourced regulators, and corporate reliance on trade secret defences.[28] Scholars have long observed that European data protection law is conceptually overburdened and structurally ill-suited to deliver transparency in such contexts.[29]

As Foucault observed, knowledge is inseparable from power: when metrics and validation standards are privately controlled, epistemic exclusion is naturalised.[30]

2.2.4 State-Enforced Enclosure

State-enforced enclosure refers to sovereign restrictions justified by national security, industrial strategy, or ideology.

Mechanisms include:

- **Data localisation laws** – requiring certain data to remain within national borders;

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[26] Cathy O'Neil, *Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy* (Penguin 2016) 21–23.

[27] European Data Protection Board, 'Guidelines on Automated individual decision-making and Profiling for the purposes of Regulation 2016/679' (WP251rev.01, adopted 3 October 2017, last revised 6 February 2018).

[28] Directive (EU) 2016/943 [2016] OJ L157/1.

[29] B-J Koops, 'The Trouble with European Data Protection Law' (2014) 4 *International Data Privacy Law* 250, 253–256.

[30] Michel Foucault, *Power/Knowledge: Selected Interviews and Other Writings 1972–1977* (Colin Gordon tr, Pantheon 1980) 98–102.

- **Pre-deployment audit mandates** – disclosure of data or models to regulators;
- **Export controls** – restricting AI models, code, or HPC hardware.

China exemplifies a comprehensive state-led regime. The Cybersecurity Law 2016 requires localisation and security review; the Data Security Law 2021 imposes a tiered classification of 'important' and 'core' data; the Personal Information Protection Law 2021 mirrors GDPR in part but subordinates privacy to state security. Article 38 PIPL obliges foreign firms to undergo CAC security assessments before cross-border transfers. The Algorithmic Recommendation Provisions 2022 require recommender systems to be registered and aligned with state-defined ideological norms (arts 7, 17).

These measures embed sovereign oversight into AI governance, limiting foreign collaboration and curbing independent auditability.

2.3 Interrelations and Overlaps

In practice, the four enclosure types rarely occur in isolation. A single AI system may be:

- protected by trade secrets (**legal enclosure**),
- accessible only via a closed API (**technical enclosure**),
- benchmarked against undisclosed proprietary datasets (**epistemic enclosure**), and
- subject to export restrictions or security reviews (**state-enforced enclosure**).

Removing one barrier will not dissolve the others. As Cohen emphasises, informational capitalism rests on layered architectures of exclusion—legal, infrastructural, and discursive—that reinforce one another and must be confronted together.[31]

[31] Julie E Cohen, *Between Truth and Power* (Oxford University Press 2019) 170-201.

2.4 Summary Table of Restricted Knowledge

Enclosure Type	Primary mechanisms of enclosure	Typical AI-sector examples	Principal risks / effects	Indicative legal or policy responses	Evidential hooks / authorities
Legal	Patents (technical contributions); copyright in code; database rights; trade secrets; contractual terms	Patent portfolios over model architectures; copyright claims over generated code; database rights in training data; NDAs over pipelines	Anticommons fragmentation; barriers to entry; conflicts between disclosure (GDPR) and secrecy (Directive 2016/943)	Targeted compulsory licensing; limits on trade secret scope where public rights engaged; open-access carve-outs for publicly funded research	EPO Guidelines G-II, 3.3 (2025); Patents Act 1977; *Alice Corp v CLS Bank* (573 US 208, 2014); Directive (EU) 2016/943
Technical	API gating; rate-limiting; DRM; closed model weights; centralised HPC	Foundation models accessible only via API; reliance on AWS, Azure, Google Cloud	Architectural lock-in; de facto standard-setting; dependence on few infrastructure providers	Interoperability and portability mandates; FRAND terms for critical inputs; public HPC investment	Lawrence Lessig, *Code: And Other Laws of Cyberspace* (1999) 6–7; cloud market concentration data (Ofcom 2023)
Epistemic	Opaque architectures; closed benchmarks; unverifiable performance metrics	Proprietary safety evaluations; secret toxicity/bias datasets; unverifiable claims of fairness	Inability to verify accuracy; weak contestability rights; accountability gap	Statutory audit rights; disclosure baselines; civil society contestation powers	GDPR arts 13–15, 22 & Recital 71; CJEU *Nowak v Data Protection Commissioner* (C-434/16); Cathy O'Neil, *Weapons of Math Destruction* (2016)
State-enforced	Data localisation; export controls; mandatory algorithm filings; ideological content rules	Cross-border data transfer restrictions; algorithm registration with CAC; HPC export bans	Curtailment of cross-border research; chilling effect on collaboration; alignment with state ideology	International data-transfer frameworks; mutual recognition of audits; safeguards for research exceptions	Cybersecurity Law (2016); Data Security Law (2021); Personal Information Protection Law (2021) art 38; Algorithmic Recommendation Provisions (2022) arts 7, 17

Table 1.
Taxonomy of Restricted Knowledge in AI Governance

Notes: The categories are analytic lenses rather than silos. Compound enclosures (e.g. trade secret protection + API gating + closed benchmarks) require layered responses that target each vector of control. The 'indicative responses' column previews Chapter 7–8 proposals and is not exhaustive.

2.5 Implications for Legal and Policy Reform

This taxonomy equips me to assess not only how enclosures arise, but also which broad categories of reform are most suited to each mode of restriction. For example:

- **Legal measures** may include compulsory licensing for critical uses, narrowing the scope of trade secret protection where safety or rights are implicated, or open-access requirements for outputs of publicly funded research.

- **Technical measures** may involve targeted subsidies for compute, investment in public infrastructure, or minimum standards for interoperability and architectural openness.

- **Epistemic measures** may include enforceable transparency benchmarks, rights of contestation for affected parties, and independent AI audit regimes with publicly accessible summaries.

- **State-level measures** may require cross-border data arrangements, mutual recognition of algorithmic audit protocols, or diplomatic engagement to address restrictive localisation policies.

Evans emphasises that developmental success depends on a 'differentiated repertoire of institutional arrangements', not reliance on a single model of state or market control.[32] Similarly, Cohen shows how exclusion in informational capitalism operates through layered architectures—legal, infrastructural, and discursive—so reforms must be calibrated to address each dimension of enclosure rather than treated in isolation.[33]

The Taxonomy of Restricted Knowledge provides such a repertoire in the AI context, enabling interventions to be proportionately matched to legal, technical, epistemic, and state-enforced forms of restriction.

3. The Third Enclosure Movement

This chapter situates my concept of the Third Enclosure Movement within the longer history of enclosure as a governance logic. It traces how proprietary control has shifted from land, to information, and now to AI's infrastructures—sustained through legal, technical, epistemic, and state-enforced mechanisms. This framing builds on the Taxonomy of Restricted Knowledge (Chapter 2) and prepares the theoretical analysis in Chapter 4.

3.1 Introduction

The Third Enclosure Movement, introduced in Chapter 1, gains analytical force when read against its predecessors. What distinguishes this phase is not simply the extension of property to a new domain, but the recursive embedding of exclusion into the infrastructures of knowledge production themselves. Unlike earlier enclosures, AI governance combines doctrinal rights, technical design, and sovereign mandates in mutually reinforcing ways. Understanding this continuity and intensification clarifies why incremental reforms often fail and why layered responses are required.

3.2 The First Enclosure Movement

Between the sixteenth and nineteenth centuries, Parliament passed successive Enclosure Acts, authorising the fencing and surveying of common fields, meadows, and forests. These measures extinguished customary rights—such as grazing and gleaning—that had sustained local economies.

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[32] Peter Evans, *Embedded Autonomy: States and Industrial Transformation* (Princeton University Press 1995) 12–16; Peter Evans, 'The New Commons vs the Second Enclosure Movement: Comments on an Emerging Agenda for Development Research' (2005) 40 *Studies in Comparative International Development* 85, 87.

[33] Julie E Cohen, *Between Truth and Power: The Legal Constructions of Informational Capitalism* (Oxford University Press 2019) 170–201.

Polanyi described this process as the 'fictitious commodification' of land, labour, and money, whereby social institutions were redefined as commodities.[34] Thompson highlighted how enclosure dismantled moral economies of reciprocity and mutual aid.[35]

The parallel with AI lies in law's role in converting collectively accessible resources into exclusive domains, justified by narratives of efficiency or modernisation. In both contexts, ownership definitions were not neutral but reshaped the very conditions of participation and control.

3.3 The Second Enclosure Movement

James Boyle coined the term 'Second Enclosure Movement' to describe the extension of property rights to informational goods such as software, databases, genetic sequences, and digital content.[36]

At the international level, the TRIPS Agreement entrenched this logic by requiring patents to be available 'for any inventions, in all fields of technology'.[37] Though AI was not explicitly mentioned, its methods and systems fall within this broad mandate. WTO dispute settlement has confirmed the binding nature of these obligations, limiting domestic regulatory autonomy—for instance, in India – Patent Protection for Pharmaceutical and Agricultural Chemical Products (WT/DS50, 1998). Scholars highlight how TRIPS was shaped by concentrated corporate lobbying: Sell describes it as 'private power through public law',[38] while Drahos and Braithwaite characterise the outcome as 'information feudalism'.[39]

Domestic measures reinforced this enclosure logic. In the UK, the Copyright, Designs and Patents Act 1988, s 9(3) created authorship rules for computer-generated works, though their application to autonomous AI remains untested. The Copyright and Rights in Databases Regulations 1997, SI 1997/3032, implementing Directive 96/9/EC [1996] OJ L77/20, introduced sui generis database rights based on investment rather than creativity—relevant to disputes over AI training datasets.

Together, these developments consolidated a rights-based paradigm for information control, laying the doctrinal foundation for today's phase of enclosure.

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[34] Karl Polanyi, *The Great Transformation: The Political and Economic Origins of Our Time* (Beacon Press 2002) 35-39.

[35] E P Thompson, *Customs in Common: Studies in Traditional Popular Culture* (The New Press 1993) 97-120.

[36] James Boyle, *The Public Domain: Enclosing the Commons of the Mind* (Yale University Press 2008) 42–54.

[37] Agreement on Trade-Related Aspects of Intellectual Property Rights, Marrakesh Agreement Establishing the WTO, Annex 1C, 1869 UNTS 299, art 27.

[38] Susan K Sell, *Private Power, Public Law: The Globalization of Intellectual Property Rights* (Cambridge University Press 2003) 85–92.

[39] Peter Drahos and John Braithwaite, *Information Feudalism: Who Owns the Knowledge Economy* (Routledge 2002) ch 2.

3.4 Knowledge Commons and Resistance

Enclosure has never gone unchallenged. In response to the expansion of IP rights during the Second Enclosure Movement, new legal, technical, and institutional mechanisms emerged to preserve or recreate common spaces.

One prominent strand of resistance to enclosure has been the open-source software movement, embodied in the GNU General Public License ('GPL'), which operationalised the principle of openness through enforceable contractual terms. Courts have upheld such licences as binding conditions of use: in *Jacobsen v Katzer* 535 F 3d 1373 (Fed Cir 2008), the Federal Circuit confirmed that breaching open-source licence terms could constitute copyright infringement rather than merely contractual breach. The broader 'knowledge commons' literature situates such initiatives as institutional efforts to sustain open access to shared resources in the face of enclosure pressures.[40]

The Creative Commons initiative extended similar logic to cultural and scientific works, enabling creators to permit re-use and remix within specified conditions.[41] These alternative licensing regimes show how doctrinal IP frameworks can be re-purposed to institutionalise sharing.

A parallel strand drew on commons theory. Elinor Ostrom demonstrated that communities can sustainably govern common-pool resources under institutional conditions such as boundary clarity, collective rule-making, and graduated sanctions.[42] Subsequent work extended her framework to intangible resources, including scientific data and open educational resources.[43]

Peter Evans highlighted the political dimension, arguing that the survival of 'new commons' depends on whether governance arrangements reinforce exclusion or foster stewardship.[44] His point is particularly relevant to AI governance, where commons-based infrastructures (e.g. open benchmarks, data trusts) remain vulnerable to corporate capture.

These movements matter not only for resisting proprietary expansion, but for modelling alternative governance logics—licensing schemes, fiduciary arrangements, and community governance—that centre participation, transparency, and accountability. Their significance for AI lies in their refusal to treat knowledge purely as a market commodity, providing institutional templates for partial commons-based reforms later developed in Chapter 7.

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[40] Brett M Frischmann, Michael J Madison and Katherine J Strandburg (eds), *Governing Knowledge Commons* (Oxford University Press 2014).

[41] Lawrence Lessig, *Free Culture: How Big Media Uses Technology and the Law to Lock Down Culture and Control Creativity* (Penguin 2004) 85–112.

[42] Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge University Press 1990) 90–101.

[43] Charlotte Hess and Elinor Ostrom (eds), *Understanding Knowledge as a Commons: From Theory to Practice* (MIT Press 2007).

[44] Peter Evans, 'The New Commons vs the Second Enclosure Movement' (2005) 40 *Studies in Comparative International Development* 85, 87.

3.5 The Third Enclosure Movement

I define the Third Enclosure Movement as the consolidation of legal, technical, epistemic, and sovereign controls over AI's core infrastructures. Its key features include:

- **Platform governance** – API-mediated access to foundation models, which restricts replication and audit.
- **Compute monopolies** – concentration of high-performance computing among a handful of providers.
- **Epistemic asymmetry** – privatisation of benchmarks and performance evaluation.
- **State-enforced restrictions** – sovereign controls over data flows and model deployment, exemplified by China's Personal Information Protection Law 2021 and the Provisions on the Management of Algorithmic Recommendations in Internet Information Services 2022, which require model registration and filing obligations with the Cyberspace Administration of China (arts 7, 17).

This phase is not simply a continuation of the Second Enclosure Movement. It alters the conditions of epistemic agency—determining who may know, how, and under what terms. Julie Cohen's account of informational sovereignties[45] and Shoshana Zuboff's analysis of surveillance capitalism[46] capture aspects of this transformation, though neither fully addresses its recursive, polycentric character.

By framing AI governance as the Third Enclosure Movement, I emphasise how legal rights, infrastructural concentration, epistemic opacity, and sovereign oversight interact to produce compound, mutually reinforcing barriers. It is this layered structure—rather than any single doctrine—that most sharply distinguishes the present phase of enclosure.

3.6 Conclusion

From common fields to datasets, the logic of enclosure has persisted: shared resources are converted into exclusive domains under the banners of innovation, efficiency, or security. What distinguishes the Third Enclosure Movement is its multidimensional structure, combining legal rights, infrastructural concentration, epistemic opacity, and sovereign control.

This complexity underlines why conventional IP reform is insufficient. Effective governance requires tools capable of addressing overlapping modalities of restriction simultaneously. The Taxonomy of Restricted Knowledge (Chapter 2) provides such a framework, and the next chapter develops its normative grounding by engaging with Rawls, Marx, Ostrom, and Foucault.

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[45] Julie E Cohen, *Between Truth and Power* (Oxford University Press 2019) ch 6.

[46] Shoshana Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (Profile Books 2019) 197–231.

4. Foundational Theoretical Frameworks

This chapter situates my argument within four influential traditions of legal and political thought—Rawlsian liberal egalitarianism, Marxist critique, Ostromian commons theory, and Foucauldian knowledge–power analysis. Each provides a distinct normative lens for assessing The Third Enclosure Movement, complementing the diagnostic taxonomy in Chapter 2 and deepening the historical framing in Chapter 3. Together, they support my preference for partial commons-based interventions as a feasible and normatively defensible regulatory path.

4.1 Rawlsian Liberal Egalitarianism

John Rawls's theory of justice evaluates social institutions against two principles: equal basic liberties for all, and inequalities permitted only where they benefit the least advantaged (the difference principle).[47]

Applied to AI governance, the difference principle requires scrutiny of whether enclosures—such as proprietary access to foundation models or monopolised compute—improve the position of those excluded from innovation. In practice, they often entrench disparities in digital capacity and economic opportunity. A pertinent example is GDPR Article 22, which restricts decisions 'based solely on automated processing' and, read with Articles 13–15 and Recital 71, requires disclosure of meaningful information. These safeguards are designed to protect individuals from epistemic asymmetries, yet their limited scope and weak enforcement reveal how enclosure regimes fall short of Rawls's demand that inequalities serve the least advantaged.[48]

Rawls's principle of fair equality of opportunity resonates similarly. Where foundational AI tools are accessible only through costly APIs or when high-performance computing resources are concentrated among a few firms, structural opportunities for research and participation are curtailed. Here, competition law offers a partial response: the Digital Markets Act imposes interoperability and data-sharing obligations on 'gatekeepers' precisely to prevent incumbents from foreclosing digital markets.[49] These measures can be read as institutional instantiations of Rawls's requirement that opportunity be genuine and not illusory.

Redistributive mechanisms—such as fiduciary data trusts, open-licensing for publicly funded AI, or shared HPC infrastructure—should therefore be understood as requirements of justice, not discretionary benevolence. Joshua Cohen extends this Rawlsian logic by arguing that background institutions must sustain both democracy and equality.[50] This reinforces the concern, developed in Chapter 9, that unregulated AI enclosures risk hollowing out both public accountability and epistemic plurality.

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[47] John Rawls, *A Theory of Justice: Revised Edition* (Harvard University Press 1999) §§ 11–17.

[48] Regulation (EU) 2016/679 (GDPR) [2016] OJ L119/1, arts 13–15, 22; Recital 71.

[49] Regulation (EU) 2022/1925 (Digital Markets Act) [2022] OJ L265/1, arts 6(4)–(9).

[50] Joshua Cohen, 'Democratic Equality' (1989) 99 *Ethics* 727, 734–738.

Rawls thus provides both a diagnostic and a normative standard: enclosure regimes that fail to satisfy his fairness principles demand structural reconfiguration rather than marginal efficiency adjustments.

4.2 Marxist Critique of Capitalist Accumulation

Karl Marx treats property relations not as neutral rules but as juridical mechanisms reproducing class domination. In *Capital, Volume I*, he identifies private property and exclusive rights as forms that consolidate capital's control over labour and production.[51] Marx's account of primitive accumulation—the violent historical process through which common resources were transformed into private property—provides the antecedent for later theories of enclosure.[52]

David Harvey reformulates this as accumulation by dispossession: new commodities are continuously created through the privatisation of previously shared resources.[53] The Third Enclosure Movement exemplifies this dynamic in AI: publicly available text, image, and audio data are appropriated to train proprietary models, later monetised through restrictive licences and paywalls.

Doctrinally, this commodification is entrenched by TRIPS Article 27, requiring WTO members to make patents available across all fields of technology, and by Directive 96/9/EC, which grants sui generis database rights based on investment rather than creativity.[54] Both expand proprietary claims over informational inputs vital to AI development.

Susan Sell documents how TRIPS itself was driven by concentrated corporate lobbying, embedding Northern dominance in global knowledge production.[55] Similar asymmetries re-emerge in AI, where models trained on globally sourced data are controlled by a small number of multinational firms.

From a Marxist standpoint, algorithmic opacity is not a technical accident but a structural device: black-box systems discipline user behaviour, constrain labour, and foreclose contestability while extracting behavioural and operational data. Marx provides few incremental prescriptions, but his framework underscores why reforms that leave commodification intact risk reproducing inequality. My preference for partial commons governance acknowledges this critique while recognising the political and economic barriers to wholesale abolition.

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[51] Karl Marx, *Capital: A Critique of Political Economy, vol I* (Fowkes B tr, Penguin Classics 1990) 344–350.

[52] Karl Marx, *Capital, vol I* (Fowkes B tr, Penguin Classics 1990) 873–876, ch 26.

[53] David Harvey, *A Brief History of Neoliberalism* (Oxford University Press 2005) 145–147; David Harvey, *The Enigma of Capital: And the Crises of Capitalism* (Oxford University Press 2010) 138–145.

[54] Directive 96/9/EC [1996] OJ L77/20, arts 7–10.

[55] Susan K Sell, *Private Power, Public Law: The Globalization of Intellectual Property Rights* (Cambridge University Press 2003) 85–92.

4.3 Ostromian Commons Governance

Elinor Ostrom challenged the assumption of the 'tragedy of the commons', demonstrating through empirical studies that communities can sustainably govern shared resources under appropriate institutional conditions.[56] Her design principles—clear boundaries, proportionality of benefits to contributions, collective-choice arrangements, monitoring, graduated sanctions, and accessible dispute resolution—have since been extended to intangible resources such as research data and open-source software.[57]

For AI governance, Ostrom's insights are not merely theoretical. They find partial codification in instruments such as the EU Data Governance Act, which establishes rules for 'data intermediation services' and 'data altruism organisations'. Articles 5–10 impose governance duties on intermediaries—including registration, transparency, and conflict-of-interest safeguards—while Recitals 24–30 frame these bodies as trust-like institutions designed to protect contributor and user interests.[58] These provisions instantiate Ostrom's principles in legal form by defining boundaries, embedding collective representation, and institutionalising accountability in data stewardship.

Legal enforceability of openness also matters. Courts have confirmed that open-source licensing conditions are binding: in *Jacobsen v Katzer* the Federal Circuit held that breaching a GPL-style licence could amount to copyright infringement, not merely breach of contract.[59] This demonstrates how commons norms can be embedded directly in positive law.

Nonetheless, two challenges complicate direct application. First, the global scale of infrastructures such as foundation models exceeds the local contexts of Ostrom's case studies. Second, technical complexity risks excluding meaningful participation unless mediated by institutions that can translate expertise into accessible governance. As Evans notes, sustainable institutional design requires 'a differentiated repertoire of arrangements' rather than reliance on any single governance model.[60]

Despite these limits, Ostrom's framework remains an essential counterpoint to enclosure logics. It provides normative and institutional design guidance for fiduciary data trusts, community-governed HPC facilities, and Commons AI pilots (Chs 7–8), embedding accountability and resisting re-enclosure by dominant private actors.

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[56] Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge University Press 1990) 88–102.

[57] Charlotte Hess and Elinor Ostrom (eds), *Understanding Knowledge as a Commons: From Theory to Practice* (MIT Press 2007); Brett M Frischmann, Michael J Madison and Katherine J Strandburg (eds), *Governing Knowledge Commons* (Oxford University Press 2014).

[58] Regulation (EU) 2022/868, arts 5–10, recs 24–30.

[59] 535 F 3d 1373 (Fed Cir 2008).

[60] Peter Evans, 'The New Commons vs the Second Enclosure Movement' (2005) 40 *Studies in Comparative International Development* 85, 87.

4.4 Foucauldian Knowledge–Power Dynamics

Michel Foucault's work demonstrates that knowledge production is inseparable from the power structures that define and legitimise it. In *Discipline and Punish*, he shows how disciplinary power operates not only through coercion but by embedding control into institutions that regulate what counts as valid knowledge.[61] In *Power/Knowledge*, he articulates this more directly, describing how 'regimes of truth' emerge from practices that normalise and privilege certain epistemic claims over others.[62]

Michel Foucault's work on discourse, surveillance, and governmentality highlights how knowledge production is inseparable from power relations.[63] Applied to AI, this shifts attention from ownership to epistemic control: who decides what counts as valid knowledge, and whose interests such definitions serve.

Trade secrecy, proprietary benchmarks, and closed evaluation protocols are not only commercial strategies but mechanisms of epistemic gatekeeping, concentrating the authority to validate system behaviour in a small number of actors. The EU AI Act reflects this tension. Although Articles 16–29 impose disclosure duties on providers of high-risk AI, Article 70(2) (allowing protection of confidential information) permits firms to withhold model details under trade secret defences.[64] Such carve-outs can neutralise transparency obligations, demonstrating how legal frameworks themselves reproduce epistemic asymmetries.

Julie Cohen extends this analysis in her account of 'informational capitalism', where governance is increasingly mediated by opaque algorithmic systems that resist democratic oversight.[65]

Three dynamics of the Third Enclosure Movement align with this Foucauldian lens:

- **Opacity as power** – legal and technical mechanisms combine to shield systems from scrutiny;
- **Epistemic gatekeeping** – only a narrow set of actors may audit or challenge system logic;
- **Discursive capture** – platforms define normative baselines ('accuracy', 'safety', 'fairness') that shape regulatory discourse itself.

Foucault's perspective warns that even well-intentioned reforms risk becoming new modes of control if they entrench procedural compliance without broadening substantive epistemic plurality.

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[61] Michel Foucault, *Discipline and Punish: The Birth of the Prison* (Alan Sheridan tr, Vintage Books 1995) 195–200.

[62] Michel Foucault, *Power/Knowledge: Selected Interviews and Other Writings 1972–1977* (Colin Gordon tr, Pantheon 1980) 98–102.

[63] Michel Foucault, *Power/Knowledge: Selected Interviews and Other Writings 1972–1977* (Colin Gordon tr, Pantheon 1980) 98–102.

[64] Regulation (EU) 2024/1689 [2024] OJ L1689/1, arts 16–29, art 70(2).

[65] Julie E Cohen, *Between Truth and Power* (Oxford University Press 2019) 170–175.

4.5 Synthesising Tensions

Although divergent in method, the four frameworks converge in their critique of exclusionary asymmetries.

- **Rawls** offers a distributive test: enclosure regimes that fail to benefit the least advantaged (difference principle, *A Theory of Justice* §§11–17) cannot be justified. In AI, this benchmark is operationalised through safeguards such as Article 22 GDPR, which constrains automated decision-making.

- **Marx** highlights commodification: intellectual property doctrines like TRIPS Article 27 and the EU Database Directive 96/9/EC treat knowledge as property, embedding structural inequality in global AI governance.

- **Ostrom** demonstrates, through empirical design principles, that collective governance of shared resources is feasible; the EU Data Governance Act (Regulation (EU) 2022/868, arts 5–10) reflects this approach in institutionalising fiduciary-style duties for data intermediaries.

- **Foucault** reminds us that even disclosure mandates may entrench power asymmetries when paired with exemptions—such as Article 70(2) of the AI Act, which allows trade secret claims to override transparency.

The divergences are equally instructive. Rawls presupposes institutional legitimacy; Marx doubts reform short of structural rupture; Ostrom is optimistic about self-governance but less attuned to global hierarchies; Foucault avoids prescriptive solutions, focusing instead on how power reproduces itself through epistemic norms.

The partial commons approach developed in this dissertation draws selectively on each: Rawls for normative fairness, Marx for attentiveness to commodification and global asymmetries, Ostrom for practical governance mechanisms, and Foucault for vigilance against recursive re-enclosure. This pluralist synthesis underpins the jurisdictional analysis in Chapter 5 and the sectoral case studies in Chapter 6, ensuring that the reform proposals in Chapters 7 and 8 remain legally grounded while sensitive to the dynamics of power and exclusion.

5. Comparative Legal Environments – UK, EU, US, and China

This chapter applies the Taxonomy of Restricted Knowledge (Chapter 2) to examine how four major jurisdictions—the United Kingdom, European Union, United States, and People's Republic of China—structure access to the intangible infrastructures of AI through their legal, technical, epistemic, and state-enforced regimes. The analysis is organised by jurisdiction, with each mapped to the taxonomy categories.

The objective is twofold. First, to identify how legal doctrines, policy logics, and institutional architectures generate specific forms of AI enclosure. Second, to locate areas of convergence and divergence, particularly

in emergent norms on transparency, competition, and sovereign data governance. This comparative lens also reveals that while each jurisdiction exhibits distinctive patterns, none is free of enclosure risks.

5.1 The UK Approach

The UK's post-Brexit regulatory trajectory blends inherited EU frameworks—most notably in data protection—with domestic intellectual property law and competition policy. The current narrative emphasises 'pro-innovation' flexibility, but structural and epistemic constraints persist.

5.1.1 Legal and Epistemic Enclosure: Copyright and Data Protection

The Copyright, Designs and Patents Act 1988 ('CDPA') is the UK's principal IP statute governing subsistence of rights in software and, increasingly, AI-generated outputs. Section 1(1)(a) defines 'literary works' to include computer programs, extending copyright to AI source code. Section 9(3) attributes authorship of a 'computer-generated work'—where 'there is no human author'—to 'the person by whom the arrangements necessary for the creation of the work are undertaken'.

Originally drafted with deterministic systems in mind, s 9(3) creates interpretive uncertainty for outputs from generative AI, including large language models ('LLMs'). No UK authority has yet resolved whether such outputs satisfy the statutory test. This uncertainty enables strategic enclosure: firms controlling AI systems may assert rights over outputs whose authorship status is unclear, shaping reuse and downstream innovation.

Data protection law in the UK is anchored in the Data Protection Act 2018 ('DPA 2018') and the UK General Data Protection Regulation ('UK GDPR'). Articles 13–15 UK GDPR require disclosure of 'meaningful information about the logic involved' in automated processing, while Article 22 restricts decisions 'based solely on automated processing' producing legal or similarly significant effects. Recital 71 clarifies that such disclosure is ancillary to existing rights; case law and ICO guidance confirm there is no stand-alone 'right to explanation'. The narrowness of remedies was underscored in *Lloyd v Google LLC* [2021] UKSC 50, where the Supreme Court rejected representative damages claims for privacy breaches, limiting collective contestation.

From the taxonomy perspective:

- **Legal enclosure** arises from broad copyright protection over code and potentially AI outputs.
- **Epistemic enclosure** is addressed in principle by data protection disclosure rights but weakened by vague statutory language, under-resourced enforcement, and trade secret claims that limit transparency.

5.1.2 Structural and Technical Enclosure: Market Dynamics and Infrastructure

The UK has no AI-specific competition statute. Structural enclosure emerges through dominant platforms integrating AI into closed ecosystems, using restrictive APIs, exclusive licensing, and vertical integration. Technical enclosure is reinforced by reliance on foreign-owned HPC infrastructure (AWS, Microsoft Azure, Google Cloud), leaving the UK dependent on external providers for critical compute capacity. This dependency limits the state's leverage in imposing openness or facilitating commons-based initiatives.

5.1.3 Reform Trajectories: Post-Brexit Divergence and Sandboxing

The National AI Strategy promotes sector-led oversight and regulatory sandboxes. A House of Lords Select Committee endorsed sandboxes but warned they risk entrenching incumbents without safeguards.[66] The Intellectual Property Office has consulted on expanding text-and-data-mining exceptions, which could ease legal enclosure for AI research but faces resistance from rights-holders.

Overall, the UK approach combines legacy IP rules, a data protection regime weakened by enforcement challenges, and an innovation-first narrative. Whether this reduces enclosure will depend on IPO reforms, sandbox safeguards, and domestic investment in compute to offset reliance on foreign infrastructure.

5.2 The EU Approach

The European Union has positioned itself as a global norm-setter in rights-based AI governance, embedding precautionary and fundamental-rights principles into a harmonised supranational framework. Against the taxonomy, EU measures engage all four enclosure categories—though not always consistently.

5.2.1 Legal and Epistemic Enclosure: GDPR and Database Rights

The General Data Protection Regulation ('GDPR') applies broadly to AI systems that process personal data. Article 22 restricts decisions based solely on automated processing with significant effects, while Articles 13–15 require disclosure of 'meaningful information about the logic involved'. As discussed in Chapter 2, the Court of Justice has interpreted disclosure principles expansively—*Nowak v Data Protection Commissioner* held that even examination scripts could constitute personal data—yet the European Data Protection Board ('EDPB') confirms these provisions do not create a free-standing 'right to explanation'.[67]

Enforcement is further constrained by uneven resourcing across national data protection authorities, coupled with corporate reliance on trade secrecy to resist disclosure. Consent, often relied upon as a lawful basis, is itself narrowly delimited: the EDPB stresses that valid consent must be freely given, specific, informed and unambiguous, and cannot be bundled with contractual necessity.[68] In practice, many AI

[66] House of Lords Select Committee on Artificial Intelligence, *AI in the UK: Ready, Willing and Able?* (HL Paper 100, 2018) 104.

[67] European Data Protection Board, 'Guidelines on Automated individual decision-making and Profiling for the purposes of Regulation 2016/679' (wp251rev.01, revised 2020).

[68] European Data Protection Board, 'Guidelines 05/2020 on Consent under Regulation 2016/679' (revised 2020).

platforms offer only adhesion-style terms, leaving users with little genuine choice and compounding epistemic enclosure through formal but ineffective compliance.

Alongside privacy law, the Database Directive (96/9/EC) grants a sui generis right to database makers who invest substantially in obtaining or presenting contents. In *British Horseracing Board Ltd v William Hill Organisation Ltd* (C-203/02) the CJEU upheld such rights even over non-personal data, illustrating how legal enclosure can arise from investment-based rather than creativity-based claims.

While Articles 13–15 GDPR aim to provide individuals with 'meaningful information' about automated processing, commentators note that these duties remain vaguely framed and inconsistently enforced across Member States. As Koops observes, European data protection law often promises more in principle than it delivers in practice, leaving structural opacity intact.[69]

5.2.2 Technical and Structural Enclosure: The Digital Markets Act

The Digital Markets Act ('DMA') targets 'gatekeepers' with entrenched market power. Articles 5–7 impose binding obligations including:

- data-use and sharing restrictions (Art 5),
- bans on self-preferencing (Art 6(5)),
- interoperability duties for messaging and ancillary services (Arts 6(4)–(7)), and
- data portability and access obligations (Art 6(9)).

By compelling interoperability and access, these measures aim to reduce technical enclosure. However, the DMA does not mandate disclosure of training datasets, model weights, or benchmarking protocols, leaving epistemic enclosure intact. Enforcement actions initiated by the European Commission in 2023–24 against designated gatekeepers under arts 5–7 illustrate both the Act's potential to curb structural enclosure and the uncertainty over whether it will materially expand openness or merely recalibrate incumbent dominance.[70]

5.2.3 Integrated AI-Specific Regulation: The Artificial Intelligence Act

The AI Act[71] is the first comprehensive horizontal AI regulation. Entering into force in July 2024, with staged obligations between 2025–27, it classifies AI systems by risk level (prohibited, high-risk, general-purpose). High-risk systems must undergo conformity assessments, risk management, and post-market monitoring. GPAI developers face documentation and systemic risk reporting duties.

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[69] Koops B-J, 'The Trouble with European Data Protection Law' (2014) 4 *International Data Privacy Law* 250, 259–260.

[70] Regulation (EU) 2022/1925, arts 5–7.

[71] Regulation (EU) 2024/1689.

From the taxonomy's perspective:

- **Legal enclosure** is moderated by harmonisation under one EU regime.
- **Epistemic enclosure** is partly mitigated by documentation and auditability.
- **Technical enclosure** is indirectly engaged for GPAI but without requirements for open weights or source code.

5.2.4 Synthesis: The EU's Enclosure Dialectic

EU governance embodies a dialectic:

- **Opening measures**: GDPR transparency, DMA interoperability, AI Act auditability.
- **Closing measures**: database rights and trade secret protections that sustain exclusion.

The EU thus provides the most integrated framework among the four jurisdictions, but its internal tensions show how rights rhetoric can coexist with structures that entrench The Third Enclosure Movement.

5.3 The US Approach

The United States exemplifies a fragmented, market-led governance model. It relies heavily on intellectual property rights, trade secrecy, and sectoral privacy law, with no comprehensive federal AI statute. In taxonomy terms, this produces extensive legal, technical, and epistemic enclosure, while state-enforced enclosure remains comparatively light.

5.3.1 Legal Enclosure: Copyright and Trade Secrets

The Copyright Act 1976, 17 USC protects 'original works of authorship', including software, provided they meet minimal creativity thresholds. The US Copyright Office has confirmed that works produced without human authorship are not registrable, emphasising that copyright subsists only where a human made the creative choices.[72] Nevertheless, AI developers can enclose outputs contractually—via licensing terms, API restrictions, or DRM—exercising effective proprietary control even where copyright does not apply.

Beyond outputs, disputes increasingly concern inputs: whether training on copyrighted works is permissible. The Copyright Office's ongoing review of generative AI training acknowledges the unsettled legal status of scraping copyrighted material and the risks of recursive enclosure through exclusive licensing of training data.[73]

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[72] US Copyright Office, *Copyright and Artificial Intelligence: Part II – Copyrightability* (2024) 7–12.

[73] US Copyright Office, *Copyright and Artificial Intelligence: Part III – Generative AI Training* (Pre-publication version, 2025) 15–20.

Trade secret law remains the principal enclosure mechanism. The Uniform Trade Secrets Act (UTSA), adopted in most states, and the Defend Trade Secrets Act 2016 (18 USC § 1836) protect model architectures, datasets, optimisation methods, and evaluation tools. Because these rights require no disclosure or registration, they allow firms to shield systems from independent scrutiny, reinforcing both technical and epistemic enclosure.

5.3.2 Epistemic and Legal Enclosure: Fragmented Privacy Law

The US lacks a general privacy statute. Instead, sector-specific frameworks govern:

- the Health Insurance Portability and Accountability Act 1996 (HIPAA),
- the Children's Online Privacy Protection Act 1998 (COPPA),
- the Fair Credit Reporting Act 1970 (FCRA), and
- the Gramm–Leach–Bliley Act 1999 (GLBA).

At the state level, the California Consumer Privacy Act 2018 (CCPA), as amended by the California Privacy Rights Act 2020 (CPRA), provides access, deletion, and opt-out rights.[74] Yet anonymised and aggregated data remain outside its scope, and the statute provides no 'right to explanation'. This patchwork sustains epistemic enclosure, as algorithmic decision-making remains opaque, while also reinforcing legal enclosure through broad contractual and proprietary claims over consumer data.

5.3.3 Structural Enclosure: Competition Law and Big Tech Dominance

The Federal Trade Commission ('FTC') enforces competition law under the Federal Trade Commission Act 1914, 15 USC §§ 41–58, the Sherman Antitrust Act 1890, 15 USC §§ 1–7, and the Clayton Antitrust Act 1914, 15 USC §§ 12–27. Recent proceedings show a growing willingness to treat control of data and digital infrastructure as an anticompetitive harm.

In *FTC v Facebook Inc* (complaint filed 2020, ongoing), the Commission alleged that Facebook's acquisitions of Instagram and WhatsApp violated Section 7 of the Clayton Act by entrenching data dominance and foreclosing rivals.[75] More recently, in *FTC v Amazon.com Inc* (complaint filed 2023, pending), the FTC charged Amazon under Section 5 of the FTC Act with exclusionary practices, including self-preferencing and leveraging third-party seller data to reinforce control over digital marketplaces.[76]

These complaints demonstrate the potential to extend competition law to structural enclosure in AI-relevant markets, but limits remain. US merger guidelines, even as revised in 2023, continue to

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[74] Cal Civ Code § 1798.100 et seq.

[75] *Federal Trade Commission v Facebook Inc* 560 F Supp 3d 1 (D DC 2021).

[76] *Federal Trade Commission v Amazon.com Inc* (Complaint filed, D DC 2023).

underemphasise data concentration and innovation harms; courts retain a strong focus on consumer price effects. The FTC and Department of Justice also face resourcing constraints, and there is no AI-specific statutory mandate comparable to the EU's Digital Markets Act 2022. Proposals for reform—such as creating an Algorithmic Accountability Bureau within the FTC or enacting a federal privacy and competition statute like the Consumer Online Privacy Rights Act (COPRA, Bill S 2968, 2019, proposed)—remain unrealised.

From the taxonomy's perspective, the US model continues to permit extensive technical enclosure (no interoperability duties), legal enclosure (broad contractual freedom), and epistemic enclosure (weak disclosure obligations), leaving structural remedies largely aspirational.

5.3.4 Synthesis: A High-Enclosure, Low-Regulation Model

In comparative perspective, the US represents:

- strong **legal enclosure** (IP, trade secrecy, contracts),
- entrenched **technical and epistemic enclosure** (weak transparency rights, proprietary benchmarking), and
- minimal **state-enforced enclosure**, with regulation framed in economic rather than ideological terms.

This creates the most permissive environment for The Third Enclosure Movement among the jurisdictions studied, with few structural checks on corporate consolidation of AI resources.

5.4 Comparative Epilogue: The Chinese Approach

China exemplifies state-enforced enclosure as defined in Chapter 2 (§ 2.2.4). Unlike the UK, EU, and US, which rely heavily on proprietary IP rights and corporate concentration, China embeds legal, technical, and epistemic restrictions directly into a state-led, security-first framework. Its system combines statutes, administrative regulation, and strategic planning.

5.4.1 State-Enforced Legal Enclosure: PIPL and Data Localisation

The *Personal Information Protection Law* 2021 (PIPL) mirrors GDPR-style principles of consent and purpose limitation but subordinates them to national security and public interest.[77] The *Cybersecurity Law* 2016 and the *Data Security Law* 2021 reinforce this framework through data localisation mandates and tiered classifications of 'core' and 'important' data. Enforcement lies primarily with the Cyberspace Administration of China (CAC), supported by the Ministry of Industry and Information Technology (MIIT) and the Standing Committee of the National People's Congress (NPCSC). Together, these bodies ensure that both domestic and foreign AI developers must comply with CAC security reviews before cross-border transfer.

[77] PIPL, arts 4, 28.

5.4.2 Technical and Epistemic Enclosure: Algorithmic Recommendation Provisions

The *Provisions on the Management of Algorithmic Recommendations in Internet Information Services* 2022 impose direct controls over algorithm design and deployment. Providers must:

- Register recommendation algorithms with the Cyberspace Administration of China (CAC),
- File algorithmic models and core parameters with regulators,
- Provide opt-out mechanisms for personalised recommendations,
- Avoid promoting content contrary to 'public order' or 'mainstream values'.

Article 7 requires outputs to promote 'positive energy' (正能量), aligning outputs with state ideological norms, while Article 17 grants CAC inspection rights over algorithmic logic, datasets, and optimisation strategies. Enforcement actions by the CAC have confirmed that filing and inspection requirements are legally binding, not merely aspirational.[78]

5.4.3 Sovereign Enclosure: Strategic and Geopolitical Dimensions

China's enclosure model is further entrenched by industrial policy. The *Development Plan For The Next Generation Of Artificial Intelligence* (2017) and the *14th Five-Year Plan for National Informatization* (2021–2025) frame AI simultaneously as a growth sector and a governance tool. These plans embed AI within long-term objectives of digital sovereignty, mandating alignment with socialist values and national security.

As Zhang observes, this amounts to a form of 'Chinese antitrust exceptionalism', in which regulatory instruments—competition law, data rules, and industrial policy—are deployed to advance state power rather than to pursue market-neutral goals.[79] This underscores how enclosure in the Chinese context cannot be separated from geopolitical strategy: it is designed to secure technological autonomy while subordinating AI outputs to state-defined ideological and developmental objectives.

Sovereign enclosure here operates not only through statutes but also through strategic planning that directs investment, prioritises domestic champions, and insulates AI from foreign regulatory reach. Evans's account of 'embedded autonomy' is instructive: developmental states can combine close ties to industry with sufficient independence to pursue national goals.[80] In the Chinese model, this autonomy is expressed not through liberal pluralism but through centralised steering, where state agencies such as the Cyberspace Administration of China (CAC) coordinate both legal compliance and industrial policy.

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[78] Provisions on the Management of Algorithmic Recommendations in Internet Information Services 2022, arts 7, 17.

[79] Angela H Zhang, *Chinese Antitrust Exceptionalism: How the Rise of China Challenges Global Regulation* (Oxford University Press 2021) 142–147.

[80] Peter Evans, *Embedded Autonomy: States and Industrial Transformation* (Princeton University Press 1995) 12–16.

From the standpoint of the taxonomy, sovereign enclosure thus operates across two layers: statutory mandates (e.g. localisation under the Cybersecurity Law, review powers under the PIPL) and strategic planning instruments that guide capital allocation and define permissible epistemic outputs. This dual structure makes China's regime more coherent and less contestable than liberal systems, which often rely on fragmented statutory and market-based levers.

5.4.4 Synthesis: The Statist Enclosure Model

China illustrates how all four forms of enclosure in the taxonomy—legal, technical, epistemic, and state-enforced—can be orchestrated simultaneously.

- Statutes (Cybersecurity, Data Security, PIPL) enclose data within borders.
- Regulations (Algorithmic Recommendation Provisions) condition algorithm design on ideological compliance.
- Policy plans (2017 AI Plan; 14th Five-Year Plan) embed AI within long-term sovereignty goals.

Unlike the fragmented US model or the rights-based EU approach, China integrates enforcement bodies (CAC, MIIT, NPCSC) with statutory and policy instruments into a coherent statist enclosure regime. This raises two implications: first, that commons-based reforms viable in liberal democracies are unlikely under China's system; and second, that global AI governance must account for sovereign strategies that deliberately entrench enclosure as a tool of state power.

5.5 Comparative Synthesis: Jurisdictions Mapped Against the Taxonomy

The four jurisdictions analysed display divergent configurations of enclosure across the taxonomy's categories (Table 2).

Taxonomy Category	United Kingdom	European Union	United States	China
Legal enclosure	CDPA 1988 s 9(3) unresolved AI authorship; UK GDPR disclosure limits	GDPR + Database Directive rights; DMA obligations	Copyright Act; UTSA/DTSA trade secrets; sectoral patchwork	Cybersecurity Law, Data Security Law, PIPL; Algorithmic Recommendation Provisions
Technical enclosure	Reliance on foreign-owned HPC (AWS, Azure, Google) restricts access	DMA interoperability rules reduce platform lock-in	Dominance of cloud/HPC providers (Amazon, Microsoft, Google)	State-led HPC and cloud infrastructure concentration
Epistemic enclosure	Transparency rights under DPA 2018/UK GDPR, weakened by trade secrets	GDPR disclosure duties vs. trade secrets; AI Act auditability	Weak privacy law; proprietary benchmarks shield systems	Algorithmic filing/audit obligations to CAC; outputs aligned with state norms

State-enforced enclosure	Limited; security/AI strategy narrative more market-led	Moderate; cross-border data rules via GDPR/AI Act	Low; limited to export controls/national security	High; sovereign strategy via Five-Year Plans, AI Development Plan

Table 2.
Comparative Mapping of Enclosure Types Across Jurisdictions (UK, EU, US, PRC)

Cross-jurisdictional observations

- **No fully 'open' regime** – each jurisdiction embeds significant enclosure, though vectors differ.
- **Divergent policy logics** – the EU integrates privacy and competition under rights-based regulation; the UK emphasises pro-innovation pragmatism while retaining EU-derived frameworks; the US relies on fragmented statutes and strong IP/trade secrecy; China orchestrates enclosure through statutory control and industrial policy.
- **Reform leverage points** –
    - UK: clarifying authorship under CDPA 1988 s 9(3) and strengthening audit rights under DPA 2018/UK GDPR.
    - EU: reconciling GDPR disclosure duties with database/IP rights; robust enforcement of DMA interoperability.
    - US: federal privacy legislation and algorithmic accountability mandates to supplement weak sectoral coverage.
    - China: limited scope for openness; international engagement could focus on cross-border data standards and algorithmic review protocols.

Implications for the taxonomy

- **Legal enclosure** dominates in the UK and US, driven by copyright, trade secrets, and database rights.
- **Technical enclosure** is acute in the US and UK through reliance on concentrated cloud/HPC providers.
- **Epistemic enclosure** is sustained across all jurisdictions but most explicitly contested in the EU, where transparency obligations coexist uneasily with trade secrecy.
- **State-enforced enclosure** is strongest in China, where CAC and related bodies combine statutory authority with strategic plans to channel AI development.

From the perspective of The Third Enclosure Movement, these findings reinforce the central thesis: enclosure is a structural feature of AI governance globally, albeit manifested through distinct institutional architectures. Effective reform therefore requires jurisdiction-specific strategies while recognising transnational interdependence. As Twining emphasises, legal theory itself must adapt to the conditions of globalisation, where overlapping transnational regimes and hybrid governance instruments influence outcomes as much as domestic law.[81]

6. Sector-Specific Manifestations of AI Enclosures

This chapter applies the Taxonomy of Restricted Knowledge (Chapter 2) to three high-impact sectors: healthcare and pharmaceuticals, agritech and food security, and consumer AI. It shows how enclosure mechanisms manifest differently across domains and assesses where partial commons-based reforms could counter them. The case studies demonstrate both the risks of sector-specific exclusion and the potential of targeted, institutionally grounded remedies.

6.1 Healthcare and Pharmaceuticals

AI systems in healthcare rely heavily on proprietary datasets—electronic health records, imaging archives, and genomic data—acquired through exclusive licensing or internal generation. Diagnostic algorithms built on these closed resources typically exclude independent replication or oversight.

In the UK and EU, patient-level data are primarily governed by the Data Protection Act 2018 ('DPA 2018') and the UK GDPR, and by the GDPR itself in the EU. Article 22 restricts decisions based solely on automated processing that have significant effects; Articles 13–15, read with Recital 71, require disclosure of 'meaningful information about the logic involved'. As noted in Chapter 5, CJEU jurisprudence and European Data Protection Board guidance confirm there is no autonomous 'right to explanation'.

From the taxonomy's perspective:

- **Legal enclosure** arises through intellectual property over software and exclusive licensing of datasets.
- **Epistemic enclosure** is reinforced by trade secrecy and privacy law, limiting scrutiny of algorithm design and validation.

Drug and device development costs further illustrate these pressures: DiMasi et al estimate average out-of-pocket costs for a new approved drug at over $1.3 billion.[82] Clinical trial outcomes and AI-assisted modelling are frequently shielded as trade secrets under UK common law, the US Uniform Trade Secrets

[81] William Twining, *Globalisation and Legal Theory* (Cambridge University Press 2000) 45–49.

[82] Joseph A DiMasi and others, 'Innovation in the Pharmaceutical Industry: New Estimates of R&D Costs' (2016) 47 *J Health Econ* 20, 24–28.

Act, or the EU Trade Secrets Directive.[83] This produces epistemic enclosure: regulators and researchers may be unable to verify claims, assess bias, or replicate findings—especially in low-resource settings with limited bargaining power.

Liability frameworks reinforce these concerns. In the EU, the Product Liability Directive[84] treats defective software as a product capable of causing compensable harm; in the UK, parallel liability arises under the Consumer Protection Act 1987. Where AI diagnostic systems malfunction, opacity over data and design can frustrate proof of defect, weakening patients' ability to claim redress.

The public health consequences are clear. AI systems with equitable potential—such as tuberculosis screening in underserved regions—remain locked behind licensing barriers. Regulators, lacking independent datasets, risk over-reliance on industry-supplied metrics. In Rawlsian terms (§ 4.1), such enclosures fail the difference principle by worsening the position of the least advantaged

Countermeasures include:

- Data trusts for clinical datasets, embedding fiduciary duties to patients and public health bodies;
- Open licensing for outputs of publicly funded trials;
- Mandatory audit rights for high-risk AI under GDPR Article 22, subject to narrowly defined trade secret exceptions.

These measures align with Ostrom's governance principles (§ 4.3), establishing boundaries, proportionality, and accountability mechanisms.

6.2 Agritech and Food Security

Agricultural innovation has long been shaped by enclosure logics, from seed patenting to proprietary biotechnology. In the AI era these dynamics are digitally redeployed: farmers may own their land, but informational representations of yields, soils, or pest patterns are captured and enclosed within proprietary platforms. Control increasingly lies not with producers but with those who manage data infrastructures.

6.2.1 Precision Agriculture and Sensor-Based Enclosures

Precision farming uses satellite imagery, drones, and sensor networks to generate high-resolution agronomic data. Although farmers generate this data, ownership and access typically rest with agribusiness firms or technology providers storing it in proprietary cloud environments.

- **Technical enclosure** arises where access is tied to subscription dashboards or APIs that block export.

[83] Directive (EU) 2016/943 [2016] OJ L157/1.

[84] Council Directive 85/374/EEC on liability for defective products [1985] OJ L210/29, as amended.

- **Legal enclosure** follows from adhesion contracts that assign data rights to providers and routinely limit liability for service failure. Under UK law, such clauses may be subject to the Consumer Rights Act 2015, which prohibits unfair terms in standard-form contracts, while in the EU, the Unfair Contract Terms Directive 93/13/EEC provides similar scrutiny.

This creates a divergence between physical and informational ownership: farmers retain their land but must purchase access to the datasets describing it. GDPR safeguards (see Ch 5) apply only where personal data are involved, leaving most agronomic data outside privacy law.

6.2.2 Re-deployment of Seed Patent Logic

The enclosure logic familiar from biotechnology now reappears in digital form. Kloppenburg describes this as 're-purposing the master's tools', with proprietary control once applied to seeds now extended to agronomic data and decision-support platforms.[85] Vandana Shiva makes a parallel critique in her account of biopiracy, where traditional biodiversity and indigenous knowledge were appropriated through IP regimes that converted collective resources into corporate assets.[86]

Farmers in the digital agritech economy often:

- possess only conditional use rights to platform recommendations;
- lack contractual rights to inspect predictive models;
- face barriers to transferring operational data to competing systems.

The result is epistemic enclosure: predictive logic shaping planting or harvesting remains opaque even to the farmer whose land produced the data. Interoperability is rare, locking producers into vertically integrated ecosystems. Liability risks are compounded by adhesion contracts that disclaim provider responsibility for inaccurate or harmful recommendations.

6.2.3 Fragility of Commons-Based Initiatives

Countervailing efforts include public weather databases, community sensor networks, and open agronomic models. These instantiate Ostrom's principles (§ 4.3) of shared governance but face persistent vulnerabilities:

- high infrastructure costs driving reliance on private sponsorship;
- fragmented standards in data formats, licensing, and governance;

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[85] Jack Kloppenburg, 'Re-purposing the Master's Tools: The Open Source Seed Initiative and the Struggle for Seed Sovereignty' (2014) 41(6) *Journal of Peasant Studies* 1225, 1230–1232.

[86] Vandana Shiva, *Biopiracy: The Plunder of Nature and Knowledge* (South End Press 1997) 22–30.

- geopolitical restrictions on agricultural data flows.

Without statutory carve-outs recognising agricultural data as a public-interest resource, or interoperability mandates compelling platform openness, such initiatives risk eventual absorption into proprietary ecosystems.

Sectoral synthesis: Agritech demonstrates how technical and legal enclosures converge to restrict farmers' informational sovereignty. The analogy to seed patenting highlights continuity of enclosure logics, while the fragility of commons initiatives underscores the need for binding governance structures. Within the taxonomy (Ch 2), legal and technical enclosures dominate, producing epistemic opacity and contractual asymmetries that erode farmers' autonomy.

6.3 Consumer-Facing AI

Consumer-facing AI models illustrate how enclosure dynamics play out directly in public digital markets. Unlike healthcare or agritech, these systems are visible to end-users, making their governance trajectories particularly instructive.

6.3.1 The OpenAI Pivot: From Openness to Enclosure

OpenAI's trajectory exemplifies a sequential enclosure logic. GPT-2 (2019) was released with model weights and documentation, aligning with open-science norms.[87] By 2020, GPT-3 was made accessible only via API, and GPT-4 (2023) followed the same model, limiting access to OpenAI's own servers.[88]

- **Legal enclosure**: restrictive API terms prohibit replication and certain downstream uses.
- **Technical enclosure**: model weights and training data are withheld; inference runs only on OpenAI servers.
- **Epistemic enclosure**: benchmarks and interpretability tools remain unpublished, preventing independent scrutiny.

Regulatory safeguards offer only partial counterweights. Under the GDPR, Articles 13–15 and 22 apply where personal data are processed (see Ch 5), but providers routinely invoke trade secret protection—particularly under the US Defend Trade Secrets Act 2016 (18 USC § 1836)—to limit disclosures.

Liability exposure arises under consumer protection and product liability frameworks. Misleading or unsafe outputs may be actionable under the EU Unfair Commercial Practices Directive 2005/29/EC or the US Federal Trade Commission Act 1914, 15 USC §§ 41–58. Where AI outputs cause material harm, providers

[87] OpenAI, 'GPT-2: 1.5B release' (OpenAI, 5 November 2019) <https://openai.com/index/gpt-2-1-5b-release/> accessed 23 August 2025.

[88] OpenAI, 'OpenAI API' (OpenAI, 11 June 2020) <https://openai.com/index/openai-api/> accessed 23 August 2025.

could also face claims under the EU Product Liability Directive 85/374/EEC (as amended) or the UK Consumer Protection Act 1987, which both treat defective software as a product capable of giving rise to strict liability. Yet OpenAI's standard-form contractual disclaimers typically attempt to shift risks to end-users. The enforceability of such clauses is doubtful where consumer rights are non-derogable or where harm from foreseeable misrepresentation or defect cannot lawfully be excluded.

6.3.2 Hugging Face and EleutherAI: Partial Commons Countermodels

By contrast, Hugging Face's Transformers library and EleutherAI's open models (GPT-Neo, GPT-J, GPT-NeoX) attempt to instantiate commons logics (documented in their respective technical releases):[89]

- open licences reduce legal enclosure;
- transparent benchmarking enables reproducibility;
- community governance supports collaborative curation.

Yet their sustainability is fragile. Hugging Face's venture-capital funding may generate enclosure pressures, while EleutherAI's volunteer base lacks the compute resources of corporate labs. Moreover, openly released models can be re-enclosed when incorporated into proprietary applications without reciprocal openness obligations.

6.3.3 Risks of Re-enclosure and Public Funding

Many leading consumer AI systems draw on publicly funded datasets, architectures, and subsidised compute. Without statutory openness conditions, these outputs risk privatisation—an enclosure of public investment. Countermeasures include:

- embedding openness clauses in public grants and procurement contracts;
- requiring interoperable licences for publicly funded models;
- conditioning state procurement on compliance with commons governance standards.

Sectoral synthesis

Consumer AI demonstrates the full spectrum of enclosures:

- **legal and technical** via licensing and API gating,

[89] Thomas Wolf and others, 'HuggingFace's Transformers: State-of-the-Art Natural Language Processing' (arXiv preprint, v5, 14 July 2020) arXiv:1910.03771 [cs.CL] https://doi.org/10.48550/arXiv.1910.03771 accessed 27 August 2025; Sid Black and others, 'GPT-NeoX-20B: An Open-Source Autoregressive Language Model' (arXiv preprint, 14 April 2022) arXiv:2204.06745 [cs.CL] https://doi.org/10.48550/arXiv.2204.06745> accessed 28 August 2025.

- **epistemic** via opacity of benchmarks, and
- **state-enforced** where cross-border deployment is regulated.

OpenAI, Hugging Face, and EleutherAI exemplify divergent trajectories: enclosure, fragile commons, and risks of downstream appropriation. The lesson for reform is twofold: commons initiatives require enforceable openness obligations, and liability regimes must prevent providers from externalising risk onto consumers through unfair disclaimers.

6.4 Case Study Synthesis

Table 3 summarises how the four enclosure categories from the taxonomy manifest across healthcare, agritech, and consumer-facing AI. The case studies confirm three broad lessons:

1. **Enclosures are sector-specific**: healthcare is dominated by legal and epistemic barriers; agritech by technical and contractual controls; consumer AI by the full spectrum of enclosure types.
2. **Commons initiatives exist but are fragile**: data trusts, cooperative pooling, and open-source models lack statutory protection and are prone to re-enclosure.
3. **Reform must be tailored**: compulsory licensing and audit rights suit healthcare; interoperability mandates and public-interest carve-outs are needed in agritech; procurement-linked openness is most effective for consumer AI.

Sector	Legal Enclosure	Technical Enclosure	Epistemic Enclosure	State-Enforced Enclosure
Healthcare	Exclusive clinical datasets; IP over software; trade secrecy in trials; product liability under consumer/medical device law	Closed datasets; reliance on proprietary diagnostic platforms	Opacity in algorithm design and validation; limited disclosure under GDPR/UK GDPR	Privacy and data protection regulation restrict auditability but do not guarantee explanation rights
Agritech	Adhesion contracts assigning rights to providers; limited recourse for farmers under contract law	Cloud-based data storage; proprietary dashboards; non-interoperable APIs	Opaque predictive logic; proprietary agronomic models; lack of interoperability	Few direct rules, but cross-border agricultural data sometimes restricted for food security
Consumer AI	Restrictive licensing; API gating; disclaimers limiting liability; potential tort/consumer claims for misleading outputs	API-only access; model weights withheld; compute centralisation	Benchmarks, datasets, and interpretability tools unpublished, blocking independent scrutiny	Cross-border AI deployment subject to export control and sovereign data regimes

Table 3.
Taxonomy Manifestations Across Sectoral Case Studies (Healthcare, Agritech, Consumer AI)

7. Radical Shift vs Partial Commons

This chapter contrasts two strategies for dismantling AI-related enclosures. Abolitionism calls for dismantling proprietary regimes such as intellectual property and trade secrecy, while partial commons reform reconfigures them through carve-outs, collective governance, and targeted public investment.

Abolitionist critiques, often rooted in Marxist political economy, expose the inequities of enclosure but face entrenched political and treaty-based resistance. Partial commons approaches, by contrast, are more feasible within existing legal architectures if coupled with enforceable openness obligations.

7.1 Arguments for Radical Overhaul

7.1.1 Abolitionist Positions and the Marxist Critique of IP

Radical critiques of intellectual property ('IP') often draw on Marxist political economy, which views property not as a neutral incentive mechanism but as a juridical tool for consolidating capital and reproducing class hierarchies.[90] From this perspective, privatisation of data, algorithms, and compute primarily converts collectively produced epistemic goods into rent-extracting assets.

As discussed in Chapter 3 (§ 3.3), the Agreement on Trade-Related Aspects of Intellectual Property Rights ('TRIPS') exemplifies this enclosure logic. Drahos describes the 'club model' of IP treaty-making—dominated by the US and multinational firms—as structurally exclusionary, constraining regulatory autonomy of other states.[91] Abolitionists therefore treat dismantling patents, copyright, database rights, and trade secrecy as a prerequisite for democratising knowledge governance.

Even in the United States—often portrayed as the paradigmatic IP-exporting jurisdiction—copyright protection at the training stage of AI models remains unsettled. The US Copyright Office has confirmed that fully machine-generated outputs are not registrable, but its recent reports recognise unresolved questions around the use of copyrighted works in training datasets.[92] This continuing doctrinal uncertainty illustrates why abolitionist critiques gain traction: if proprietary frameworks cannot clearly demarcate authorship or training use, then calls to dismantle IP altogether reflect frustration with systemic indeterminacy as much as with distributive inequity.

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[90] Karl Marx, *Capital: A Critique of Political Economy, vol I* (Fowkes B tr, Penguin Classics 1990) 345–350.

[91] Peter Drahos, *Information Feudalism* (Routledge 2002) ch 2.

[92] US Copyright Office, *Copyright and Artificial Intelligence: Part II – Copyrightability* (Report, 2024) <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-2-Copyrightability-Report.pdf> accessed 28 August 2025; US Copyright Office, *Copyright and Artificial Intelligence: Part III – Generative AI Training* (Pre-publication version, 2025) <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-3-Generative-AI-Training-Report-Pre-Publication-Version.pdf> accessed 28 August 2025.

7.1.2 Structural Critique of TRIPS and Global IP Enforcement

TRIPS requires patents to be available 'for any inventions, in all fields of technology',[93] a provision that disproportionately benefits IP-exporting states and restricts the industrial policy options of the Global South. In AI, models trained on globally sourced datasets—often scraped from the open web—can be patented or licensed exclusively by Northern firms. By contrast, researchers in the South may face prohibitive licensing fees even when their local data form part of the training corpus.

The treaty's enforcement structure compounds these asymmetries. WTO dispute settlement has consistently treated TRIPS obligations as binding, limiting domestic experimentation. For example, India's attempt to maintain a transitional patent regime was struck down in India – Patent Protection for Pharmaceutical and Agricultural Chemical Products (WT/DS50, 1998). National measures invoking TRIPS flexibilities—such as compulsory licensing under arts 30–31—are legally available but politically risky, often challenged as violations.

Baker highlights that even where flexibilities exist on paper, their effective use in non-producing countries depends on political will, bargaining capacity, and institutional support.[94] India's first compulsory licence in *Natco v Bayer* (2012) demonstrates how such flexibilities can be operationalised domestically, but also how quickly these efforts attract political and corporate resistance.[95]

At the transnational level, Sell shows how the Access to Medicines Campaign mobilised NGOs, developing states, and civil society to reframe IP enforcement as a public health issue, securing concessions such as the Doha Declaration.[96] Yet, despite these victories, the structural asymmetry of TRIPS enforcement remained largely intact.

7.1.3 Political Infeasibility and the Reform Impasse

Despite their analytical force, abolitionist strategies face formidable obstacles:

- **Treaty entrenchment**: TRIPS is embedded in the WTO framework; amendment requires improbable consensus, and dispute settlement practice entrenches obligations.
- **Corporate lobbying**: multinational firms invest heavily in defending global IP norms.
- **Diplomatic leverage**: IP-exporting states frequently tie trade concessions or aid to strong IP enforcement.

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[93] TRIPS, art 27.

[94] Brook K Baker, *Processes and Issues for Improving Access to Medicines: Willingness and Capacity to Utilise TRIPS Flexibilities in Non-Producing Countries* (DFID Health Systems Resource Centre 2004) 12–14.

[95] *Natco Pharma Ltd v Bayer Corporation Compulsory Licence Application No 1 of 2011* (Controller of Patents, 9 March 2012).

[96] Susan K Sell, 'TRIPS and the Access to Medicines Campaign' (2002) 20(3) *Wisconsin International Law Journal* 481, 485–490.

Sell's account of the Access to Medicines Campaign shows that even when civil society and developing states successfully reframed IP enforcement as a public health issue—securing the Doha Declaration—these gains were largely confined to interpretive flexibilities and did not alter the structural asymmetry of the treaty.[97]

As Tirole notes, without alternative mechanisms—such as innovation prizes or revenue-sharing schemes—abolition risks leaving an 'innovation vacuum' in capital-intensive sectors.[98] This is acute for AI, where infrastructure and talent costs are extremely high.

For these reasons, abolitionism may be better understood as a diagnostic critique of enclosure logics than as a viable short-term reform strategy. The next section turns to partial commons-based reform as a more politically feasible route for dismantling enclosures in ways sensitive to the political economy outlined in Chapters 5–6.

7.2 Partial Commons-Based Approaches

Partial commons reforms aim to reconfigure IP and proprietary regimes without dismantling them. They carve out spaces for open access, collective governance, and equitable participation, mapped directly to the four forms of enclosure in the Taxonomy of Restricted Knowledge (Ch 2). Unlike abolitionism, these approaches work within existing legal architectures but deliberately rebalance rights and obligations.

7.2.1 Open Licensing Carve-Outs for Publicly Funded AI

Publicly funded AI research should not be re-enclosed by private contractors. Grant conditions can require models, datasets, and code to be released under open licences such as Creative Commons Attribution or Apache 2.0.

- **Legal enclosure** is reduced by preventing downstream exclusivity.
- **Epistemic enclosure** is weakened by requiring that model architectures, documentation, and benchmarks be open for scrutiny.

Comparative precedents:

- **EU Horizon Europe** requires open access to publications and data unless security or confidentiality applies.
- **US federal grant policy** increasingly mandates open data for funded projects (e.g., NIH Public Access Policy). These mechanisms illustrate that licensing carve-outs can be operationalised without dismantling IP law.

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[97] Susan K Sell, 'TRIPS and the Access to Medicines Campaign' (2002) 20(3) *Wisconsin International Law Journal* 481, 490–495.

[98] Jean Tirole, *Economics for the Common Good* (Rendall S tr, Princeton University Press 2017) 45–48.

7.2.2 Data Trusts Governed by Ostromian Principles

Data trusts are legal or organisational entities that steward datasets on behalf of defined beneficiaries, setting access terms, overseeing compliance, and representing the collective interests of contributors. Properly designed, they can counter multiple enclosure types:

- **Legal enclosure** – by holding rights collectively and licensing under agreed conditions;
- **Epistemic enclosure** – by embedding disclosure and audit rights in trust rules;
- **Technical enclosure** – by pooling infrastructure to ensure equitable access.

To avoid tokenism or corporate capture, data trusts should follow Elinor Ostrom's design principles for commons governance—boundary clarity, collective-choice arrangements, monitoring, graduated sanctions, and nested governance.[99] This institutional logic has been extended to knowledge resources in legal scholarship, most notably by Frischmann, Madison and Strandburg, who frame data stewardship as a form of governing knowledge commons requiring bespoke governance rules rather than generic IP or contract law defaults.[100]

For example, a national health data trust could require that any AI trained on its datasets release validation benchmarks, permit independent audits, and apply GDPR-compliant anonymisation.[101] In the UK, pilots led by the Open Data Institute and the AI Council have tested such models, while the EU Data Governance Act formalises similar arrangements by imposing governance duties on data intermediation services.[102]

From the taxonomy's perspective, data trusts:

- counter **legal enclosure** by vesting rights in fiduciary entities with public-interest duties;
- pierce **epistemic enclosure** by institutionalising auditability and structured disclosure; and
- mitigate **technical enclosure** by pooling shared infrastructure.

7.2.3 Hybrid Licensing Schemes

Hybrid approaches preserve incentives while guaranteeing baseline openness:

- **Tiered licensing**: royalty-free for non-commercial research; paid licences for commercial exploitation.

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[99] Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge University Press 1990) 90–101.

[100] Brett M Frischmann, Michael J Madison and Katherine J Strandburg (eds), *Governing Knowledge Commons* (Oxford University Press 2014).

[101] Regulation (EU) 2016/679 (General Data Protection Regulation) [2016] OJ L119/1, arts 5(1)(b)–(e).

[102] Regulation (EU) 2022/868 [2022] OJ L152/1, arts 5–10.

- **Time-bound exclusivity**: proprietary control for a fixed period (e.g. three years), then automatic reversion to commons use.

These models are familiar in other domains:

- The IEA's Clean Energy Transitions Programme (CETP) provides open-access analysis, policy advice, and capacity-building for emerging economies, operating under a Creative Commons (CC BY 4.0) license that enables policy use (even if commercial licensing models remain unspecified).[103]

- In medicines, the Medicines Patent Pool employs voluntary licences to expand access while maintaining incentives.[104]

Applied to AI, such schemes place temporal or use-based limits on legal enclosure, while embedding disclosure obligations at expiry or for non-commercial uses.

7.3 Reconciling Private Innovation Incentives

Partial commons reforms must address a practical question: how to sustain incentives for capital-intensive AI research while embedding enforceable openness obligations. Contemporary incentive systems rely heavily on exclusivity—whether through IP rights, contractual restrictions, or technical enclosure of infrastructure. Political feasibility depends on showing that alternative architectures can fund innovation without reverting to full proprietary control.

7.3.1 Innovation Prizes and Public Grant Schemes

Mechanism: Offer substantial rewards for achieving defined objectives, with outputs—models, datasets, documentation—released under open licences.

Taxonomy links: counters legal enclosure by conditioning rewards on openness; reduces epistemic enclosure by ensuring auditability.

Comparative examples:

- The Longitude Prize (UK) and XPRIZE illustrate non-proprietary mobilisation of innovation.

- EU Horizon Europe requires grant beneficiaries to provide open access to publications and data, unless specific security or confidentiality exceptions apply.[105]

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[103] IEA, *Clean Energy Transitions Programme 2024* (Annual Report, March 2025) 20.

[104] Medicines Patent Pool, *Annual Report 2024* (2025) 6, 8–9..

[105] Regulation (EU) 2021/695, arts 14–16.

- In the US, the NIH applies open-data mandates pursuant to the 2023 OSTP 'Nelson Memo', requiring federally funded research data to be publicly accessible.

- In biomedicine, advance market commitments for vaccines combine guaranteed purchases with eventual generic licensing.[106]

Scholarly grounding: Mazzucato argues that the state is often the primary risk-taker but fails to socialise returns.[107] Linking public finance to openness directly addresses this asymmetry.

7.3.2 State-Led HPC Infrastructure

Mechanism: Establish publicly funded, independently governed high-performance computing ('HPC') facilities accessible to academia, SMEs, and civil society, with embedded openness conditions.

Taxonomy links: targets technical enclosure by reducing reliance on a few corporate cloud providers; mitigates epistemic enclosure by requiring reproducibility pipelines and open benchmarks.

Comparative examples:

- EuroHPC pools EU and Member State resources to build exascale capacity; its mandate includes access for non-commercial research.[108]

- Japan's Fugaku supercomputer allocates capacity to public-interest AI projects with publication requirements.

Scholarly grounding: Bostrom warns that concentrated compute centralises existentially significant decisions.[109] State-led HPC distributes computational agency across a broader set of actors.

7.3.3 Commons AI Pilot Programmes

Mechanism: Develop sectoral AI projects under layered licences—open access for non-profit use, capped royalties for commercial deployment—with revenues reinvested into public funds or statutory trusts.

Taxonomy links: addresses legal enclosure through licensing terms; counters epistemic enclosure by embedding transparency conditions; reduces technical enclosure by providing shared infrastructure.

Comparative examples:

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[106] P Kremer and others, 'Advance Market Commitments for Vaccines' (2007) 358 *The Lancet* 1885.

[107] Mariana Mazzucato, *The Entrepreneurial State: Debunking Public vs Private Sector Myths* (Penguin 2018) 35–42.

[108] Regulation (EU) 2021/1173, arts 3–4.

[109] Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford University Press 2014) 115–120.

- The Medicines Patent Pool (MPP) uses non-exclusive licensing to expand access while reinvesting royalties into public health.[110]
- In AI, statutory frameworks already allow experimental governance: the EU AI Act recognises regulatory sandboxes encouraging supervised testing.[111]
- Open software initiatives such as the Community Earth System Model demonstrate sustained uptake when baseline code is openly released.

Bridging to political economy: These strategies show that partial commons reforms can:

- preserve innovation incentives for private actors,
- dismantle enclosures selectively, and
- embed openness through grant rules, procurement, and statutory pilot programmes.

7.4 Enclosure Types and Remedies

The four enclosure categories mapped in Chapter 2 can be paired with targeted partial commons remedies (see Table 4). This matrix highlights where abolitionist strategies are least feasible and where incremental reform has the greatest leverage. The aim is not to eliminate proprietary structures wholesale but to embed openness, transparency, and equitable access directly into legal and institutional frameworks.

Enclosure Type	Illustrative Mechanisms	Targeted Remedies	Feasibility Assessment
Legal	Patents, copyright, database rights, trade secrets	Compulsory licensing (TRIPS arts 30–31); Open licensing carve-outs for publicly funded AI (NIH Nelson Memo, Horizon Europe); Statutory data trusts	Abolition politically infeasible; partial commons feasible in grant/contract law
Technical	API gating, cloud concentration, DRM	Public HPC infrastructure (EuroHPC, Fugaku); Interoperability mandates (DMA arts 6(4)–(9)); Procurement-linked openness obligations	Abolition unrealistic; infrastructure remedies moderately feasible with investment
Epistemic	Opaque models, closed benchmarks, proprietary metrics	Mandatory audit rights (AI Act arts 16–29); Transparency clauses in procurement and grants; Civil-society contestation rights	Feasible with statutory support; risk of re-enclosure if trade secrets over-protected
State-Enforced	Data localisation, export controls, ideological mandates	Reciprocal data-transfer frameworks; International audit protocols; Commons AI pilot programmes under limited national security exceptions	Least amenable to reform; only partial commons pilots or soft-law frameworks viable

Table 4.
Enclosure Types and Partial Commons Remedies

[110] Medicines Patent Pool, *Annual Report 2024* (2025) 8–9, 54–57.

[111] Regulation (EU) 2024/1689, recital 60; art 53.

The remedies include:

- **Legal enclosure** → licensing carve-outs, compulsory licensing under TRIPS arts 30–31, and statutory data trusts.
- **Technical enclosure** → public HPC infrastructure (EuroHPC, Fugaku), interoperability mandates under the DMA, and procurement-linked openness.
- **Epistemic enclosure** → mandatory audit rights (EU AI Act arts 16–29), transparency obligations in grant contracts, and contestation rights for civil society.
- **State-enforced enclosure** → reciprocal data-transfer frameworks, limited security exceptions, and pilot schemes with international audit protocols.

8. Proposed Legal and Policy Solutions

This chapter translates the dissertation's analysis into a policy toolkit, using the Taxonomy of Restricted Knowledge (Ch 2) to target legal, technical, epistemic, and state-enforced enclosures. It groups reforms into three domains: data governance and transparency; competition and audit measures; and pilot programmes for Commons AI. The aim is not wholesale abolition of intellectual property, but partial commons-based solutions that are politically viable, normatively grounded, and resistant to re-enclosure.

8.1 Data Trusts and Algorithmic Transparency

Reforms in this first cluster target legal and epistemic enclosure by embedding fiduciary duties in data stewardship and mandating structured disclosure for high-impact AI systems. Data trusts and transparency regimes shift governance from opaque bilateral contracts to accountable, collectively governed arrangements.

8.1.1 Data Trusts and Fiduciary Responsibilities

A data trust is a legal entity that manages datasets on behalf of defined contributors and beneficiaries. Trustees owe duties of loyalty, care, and accountability comparable to those in corporate or pension law, ensuring access decisions are guided by contributors' best interests rather than short-term profit.

The theoretical foundations come from Elinor Ostrom's principles for governing commons—boundary clarity, collective-choice arrangements, monitoring, and sanctions—demonstrating that shared resources can be sustainably managed under the right institutional conditions.[112] These insights have been translated

[112] Elinor Ostrom, *Governing the Commons: The Evolution of Institutions for Collective Action* (Cambridge University Press 1990) 90–101.

into the legal domain by Frischmann, Madison and Strandburg, who conceptualise data stewardship as a form of governing knowledge commons, requiring governance rules tailored to the resource rather than defaulting to intellectual property or contract law.[113]

Existing instruments illustrate the move from theory to practice:

- The EU Data Governance Act regulates 'data intermediation services' and 'data altruism organisations', imposing registration, transparency, and conflict-of-interest obligations.[114]
- In the UK, pilots led by the Open Data Institute and the AI Council have tested trust frameworks for health and environmental datasets.
- Canada's Digital Charter Implementation Act (Bill C-27, 2022) proposed to modernise privacy law by introducing the Consumer Privacy Protection Act and the Artificial Intelligence and Data Act, embedding fiduciary duties for intermediaries. However, the bill was not enacted (Parliament was prorogued in January 2025), leaving PIPEDA unchanged.[115]

A statutory data trust regime should therefore require:

- clearly defined contributor and beneficiary boundaries;
- public-interest use conditions (e.g. open licensing, auditability);
- conflict-of-interest and independence rules; and
- regulatory oversight with investigatory and enforcement powers.

Taxonomy mapping: counters legal enclosure by vesting rights in fiduciary bodies bound by public-interest duties; pierces epistemic enclosure by institutionalising audit rights and structured disclosure.

8.1.2 Public–Private Data Sharing

Public datasets—NHS records, environmental monitoring, transport flows—are often shared with private firms under opaque terms and limited reciprocity. Minimum statutory or contractual conditions should include:

- algorithmic logs and reproducibility records;
- open licensing of outputs derived from public data;

[113] Brett M Frischmann, Michael J Madison and Katherine J Strandburg (eds), *Governing Knowledge Commons* (Oxford University Press 2014).

[114] Regulation (EU) 2022/868 [2022] OJ L152/1, arts 5–10.

[115] Bill C-27, Digital Charter Implementation Act 2022 (proposed, not enacted).

- reciprocity clauses obliging firms to return derived datasets, benchmarks, or financial support.

These conditions could be embedded in procurement contracts and grant terms, or issued as statutory guidance by regulators. The European Data Protection Board has emphasised that valid consent under the GDPR must be freely given, specific, informed and unambiguous.[116] In contexts where public datasets are shared with commercial actors, reliance on bundled or vaguely worded consents undermines this standard.

UK jurisprudence reinforces the point: in *Lloyd v Google LLC* [2021] UKSC 50, the Supreme Court declined a mass damages claim for loss of control of personal data, but confirmed that breaches of consent and transparency obligations remain actionable. This signals that reciprocity clauses are not only a governance tool against enclosure but also a way to reduce litigation risk by aligning data-sharing practices with enforceable privacy norms.

At the EU level, early enforcement of the Digital Markets Act illustrates how reciprocity requirements can be operationalised. In March 2024, the Commission opened non-compliance investigations against Alphabet, Apple, and Meta under Regulation (EU) 2022/1925, citing obligations on interoperability and data access (arts 6(4)–(9)),[117]

Taxonomy mapping: reduces legal enclosure by enforcing reciprocity and limits epistemic enclosure by mandating documentation.

8.1.3 Mandatory Disclosure for High-Impact AI

High-impact AI systems—recruitment, credit scoring, healthcare triage—require structured disclosure obligations. The EU AI Act introduces partial duties,[118] but its broad trade secret carve-outs risk undermining them.

A robust disclosure regime should require:

- Provenance of training data and validation methods;
- Demographic performance metrics (accuracy/bias across groups);
- Public-facing summaries intelligible to non-specialists.

To be effective, disclosure must:

- Restrict trade secret defences where systems have major public impact;

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[116] European Data Protection Board, 'Guidelines 05/2020 on Consent under Regulation 2016/679' (revised 2020).

[117] European Commission, 'Commission opens non-compliance investigations against Alphabet, Apple and Meta under the Digital Markets Act' (Press release, 25 March 2024); The investigations remain ongoing at the time of writing; no final determinations of infringement have yet been made.

[118] Regulation (EU) 2024/1689, arts 16–29.

- Require disclosure to independent auditors with statutory access;
- Permit civil society or accredited data trusts to challenge incomplete or misleading information.

Taxonomy mapping: directly pierces epistemic enclosure by enabling scrutiny, and weakens legal enclosure where proprietary claims shield public-impact systems.

8.2 Competition Law Tools and Mandatory Audits

This second cluster addresses **structural and epistemic enclosure**. While § 8.1 focused on fiduciary governance and disclosure, here the emphasis is on rebalancing market power through **antitrust enforcement, mandatory audits**, and **merger review**.

8.2.1 Antitrust Enforcement Against Data Monopolies

Dominance in AI markets often stems from exclusive control over datasets, APIs, and compute infrastructure. Traditional competition law has been slow to treat data dominance as an anticompetitive harm.

Reform directions include:

- Recognising **data concentration** as a competition factor, not just prices/output.
- Imposing **FRAND-based data-sharing** obligations in merger settlements.
- Penalising unjustified refusals to supply training datasets or APIs where competition is foreclosed.

Illustrative tools:

- The Digital Markets Act imposes interoperability duties, bans self-preferencing, and mandates business-user access to data.[119]
- In the US, the FTC has pursued exclusionary conduct cases (e.g. *FTC v Facebook*; *FTC v Amazon*, 2023). Extending such reasoning to AI-critical assets would directly counter technical enclosure.

Taxonomy mapping: weakens technical enclosure by breaking platform lock-in, and mitigates legal enclosure where restrictive contracts prevent rivals' access.

8.2.2 Algorithmic Audits and Oversight Registries

Mandatory audits provide a counterweight to epistemic enclosure, especially where structural remedies like divestiture are infeasible.

[119] Regulation (EU) 2022/1925, arts 6(4)–(9).

Audits of high-risk AI should cover:

- Fairness and non-discrimination;
- Data lineage and consent provenance;
- Reproducibility under consistent inputs;
- Appeal and contestation mechanisms.

Regulators could maintain **oversight registries** requiring disclosure of system purposes, risk classification, and audit summaries. This mirrors proposals in the Algorithmic Accountability Act (US draft S 2968, 2019).[120]

For credibility, audit regimes must:

- Guarantee auditor independence;
- Grant regulators powers to halt non-compliant systems;
- Require transparency of audit methods and results, subject only to narrow security exceptions.

Taxonomy mapping: primarily tackles epistemic enclosure, with spillover into state-enforced enclosure where registries support cross-border cooperation.

8.2.3 Merger Review and Data Openness

Merger control offers a forward-looking tool to prevent entrenched enclosures. Acquisitions consolidating unique datasets, models, or compute capacity should face enhanced scrutiny.

Reform elements:

- Expand tests to include innovation harms and data concentration.
- Require FRAND data-sharing commitments as conditions for clearance.
- Impose interoperability mandates for APIs/interfaces, echoing the EU's Microsoft server remedy.

Taxonomy mapping: reduces technical enclosure by preventing exclusive control of infrastructure, and curbs legal enclosure where proprietary rights block rivals' access.

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[120] Algorithmic Accountability Act of 2019, S 2231, 116th Cong, 1st Sess (proposed, not enacted) <https://www.congress.gov/bill/116th-congress/house-bill/2231> accessed 28 August 2025.

8.3 Pilot Schemes for 'Commons AI'

Where doctrinal reform (§ 8.1) and competition law measures (§ 8.2) cannot fully resolve enclosure, pilot programmes offer a way to test alternative governance architectures in practice. These schemes demonstrate how openness can be operationalised without abandoning innovation incentives, generating empirical evidence for scalability and resilience against re-enclosure.

8.3.1 Government–Philanthropy Partnerships

Commons AI projects can be advanced through partnerships between governments, philanthropic funders, and public research bodies. Philanthropic actors such as the Wellcome Trust, Open Philanthropy, and the Gates Foundation already support frontier projects in health, education, and climate modelling. Governments can complement this by:

- providing secure public HPC environments;
- granting regulatory sandbox status;
- issuing procurement guarantees or challenge prizes.

Partnership contracts must embed safeguards against re-enclosure, including:

- open licensing of outputs;
- audit and transparency conditions;
- sunset clauses mandating full release of models and datasets after a fixed period.

A close analogue is the Medicines Patent Pool, which expands access to essential medicines via non-exclusive licensing while reinvesting royalties in public health.[121]

8.3.2 Public Funding for Open-Access Compute and Models

A critical bottleneck for Commons AI is compute capacity. Public investment in high-performance computing ('HPC') facilities can:

- allocate compute credits to public-interest projects conditional on open release;
- require outputs to be licensed under Creative Commons or equivalent;
- reserve advanced compute for projects with strong dissemination guarantees.

[121] Medicines Patent Pool, *Annual Report 2024* (MPP June 2025) <https://medicinespatentpool.org/uploads/2025/06/MPP_AR_2024_Final.pdf> accessed 27 August 2025, 6–13.

This mirrors the Bermuda Principles adopted in the Human Genome Project, under which genome data generated with public funds had to be released within 24 hours into public repositories.[122]

From the taxonomy perspective, this directly counters technical enclosure by distributing compute, and indirectly addresses epistemic enclosure by ensuring models trained with public resources remain auditable.

8.3.3 Evaluation Metrics: Equity, Adoption, Sustainability

To avoid tokenism or corporate capture, pilot programmes must be subject to rigorous, transparent evaluation. Suggested criteria include:

- **Equity** – inclusion of marginalised groups, non-dominant languages, and Global South participation;
- **Adoption** – uptake by SMEs, universities, and public agencies;
- **Sustainability** – ability to operate without perpetual subsidy, e.g. through capped-royalty licensing or FRAND-based support tiers.

Independent review panels (academics, regulators, civil society) should oversee these metrics, publishing results to inform future legal and institutional reforms.

Conclusion to Chapter 8

The measures proposed in this chapter offer a partial commons toolkit for resisting the Third Enclosure Movement. Data trusts and disclosure regimes counter legal and epistemic enclosure; competition and audit mechanisms address structural concentration; and Commons AI pilots test governance architectures resilient to re-enclosure.

Commons AI pilots are not peripheral experiments but proofs of concept. Properly designed, they demonstrate that scalable AI ecosystems can exist outside proprietary enclosures while supplying the empirical basis for legislative and international norm-setting. By combining open licensing, shared infrastructure, and evaluation frameworks, they help institutionalise openness as a legal condition of public and philanthropic funding, rather than a matter of voluntary practice.

These reforms do not abolish proprietary regimes but incrementally re-balance rights and obligations, embedding openness, accountability, and equity into AI's governance DNA. This approach avoids the political infeasibility of abolition while creating structural pathways for reform. Chapter 9 now draws these

[122] National Human Genome Research Institute, 'Report of the International Strategy Meeting on Human Genome Sequencing' (Bermuda, 25–28 February 1996) <https://www.genome.gov/sites/default/files/archiespace2024-01/Bermuda%20Feb%201996%20Meeting.pdf> accessed 24 August 2025; National Human Genome Research Institute, 'Human Genome Project Fact Sheet' (13 June 2024) <https://www.genome.gov/about-genomics/educational-resources/fact-sheets/human-genome-project> accessed 24 August 2025.

strands together, answering the central research question and situating partial commons strategies within the wider challenge of safeguarding epistemic legitimacy and democratic agency.

9. Conclusion and Future Directions

This chapter synthesises the dissertation's findings and returns to the central research question. It concludes that wholesale abolition of intellectual property is neither institutionally feasible nor legally possible under the current international order, given the TRIPS Agreement's treaty entrenchment and its enforcement through WTO dispute settlement. Instead, partial commons-based reforms—open licensing carve-outs, fiduciary data trusts, disclosure mandates, competition remedies, and procurement-linked openness—offer a legally realistic, treaty-compliant path. These measures balance innovation incentives with enforceable obligations for transparency, auditability, and equitable access.

9.1 Answering the Central Question: Applying the Taxonomy

This dissertation has made two conceptual contributions. First, it introduced the Third Enclosure Movement as a framework for understanding how AI governance reproduces historical exclusionary logics across data, models, and infrastructure. Second, it developed the Taxonomy of Restricted Knowledge (Chapter 2), distinguishing legal, technical, epistemic, and state-enforced enclosures.

Applied to the UK, EU, US, and China (Chapter 5), and to healthcare, agritech, and consumer AI (Chapter 6), the taxonomy demonstrates that different types of enclosure demand different legal responses:

- **Legal enclosure** (copyright, database rights, trade secrecy) can be countered through open licensing of publicly funded outputs and compulsory disclosure.
- **Technical enclosure** (API gating, compute monopolies) requires public investment in high-performance computing and interoperability mandates.
- **Epistemic enclosure** (trade-secret defences, closed benchmarking) calls for statutory audit rights and transparency obligations.
- **State-enforced enclosure** (China's localisation and algorithmic regulation) requires reflexive strategies that engage with sovereignty while preserving cross-border accountability.

The conclusion is clear: abolition of intellectual property is neither feasible under TRIPS nor sufficient in practice. WTO dispute settlement has consistently treated TRIPS obligations as binding, limiting scope for domestic experimentation (e.g. *India – Patent Protection for Pharmaceutical and Agricultural Chemical Products* (WT/DS50, 1998)). As Baker shows, even where TRIPS flexibilities such as compulsory licensing are legally available, their effective use depends on political will and institutional capacity rarely present in non-

producing countries.[123] At the global level, Sell demonstrates that the Access to Medicines Campaign achieved only limited interpretive gains, leaving the treaty's core asymmetries intact.[124]

The analysis confirms that abolitionist critiques expose real doctrinal and distributive tensions, but that partial commons reforms remain the only legally feasible route under current treaty and domestic architectures. Even in the United States, long regarded as the archetype of strong IP enforcement, the Copyright Office acknowledges that the legality of training data use in generative AI remains unsettled.[125]

Partial commons reforms therefore provide a more workable path. They embed contestability and openness into existing legal architectures—through open licensing carve-outs, fiduciary data trusts, interoperability mandates, and audit rights—without dismantling innovation incentives.

9.2 Limitations and Further Research

This study has focused on four high-influence jurisdictions—the UK, EU, US, and China—while leaving aside the Global South. In many countries, AI development proceeds under sharper constraints: limited compute access, structural dependency on foreign platforms, and asymmetric data extraction.[126] Future research should test whether weaker IP regimes or regional alliances (e.g. India, Brazil, Kenya, Indonesia) permit more ambitious Commons AI experiments.

Sectoral coverage was also selective. While healthcare, agritech, and consumer AI were examined, other critical domains—finance, education, and transport—require analysis. Key tasks include: mapping data monopolies and secrecy claims; analysing case law on AI-related IP; and assessing open-access pilot outcomes.

Methodological enrichment is needed. Doctrinal and comparative analysis should be complemented by participatory and ethnographic approaches, embedding affected communities in governance design. This aligns with Rawls's fairness and Ostrom's commons principles (Ch 4). Examples include participatory audits, user-focused ethnographies, and behavioural studies of trust in AI oversight.

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[123] Brook K Baker, *Processes and Issues for Improving Access to Medicines: Willingness and Capacity to Utilise TRIPS Flexibilities in Non-Producing Countries* (DFID Health Systems Resource Centre 2004) 12–14.

[124] Susan K Sell, 'TRIPS and the Access to Medicines Campaign' (2002) 20(3) *Wisconsin International Law Journal* 481, 490–495.

[125] US Copyright Office, *Copyright and Artificial Intelligence: Part III – Generative AI Training* (Pre-publication version, 2025) <https://www.copyright.gov/ai/Copyright-and-Artificial-Intelligence-Part-3-Generative-AI-Training-Report-Pre-Publication-Version.pdf> accessed 28 August 2025.

[126] TRIPS, art 27.

9.3 Implications for the Post-Truth Era

9.3.1 AI as epistemic infrastructure

As set out in Chapter 4, AI systems now function as epistemic infrastructures: they determine what counts as valid knowledge and how it is verified. When enclosed through trade secrets, closed benchmarks, or sovereign secrecy rules, they limit contestability and reduce epistemic plurality. Existing safeguards under the GDPR (arts 13–15, art 22; Recital 71) mitigate these risks only partially, as they do not establish a free-standing right of explanation and remain narrow in scope. The result is structural distortion of informational environments. Zuboff characterises this as the normalisation of behavioural extraction under surveillance capitalism.[127]

9.3.2 Power–Knowledge Asymmetries and Regulatory Capture

The Third Enclosure Movement consolidates epistemic authority in a small set of corporate and sovereign actors. When regulators allow trade secret defences to override disclosure obligations, or defer excessively to proprietary claims, governance risks regulatory capture. Julie Cohen warns that informational capitalism operates through 'layered architectures' in which exclusion is infrastructural as well as legal.[128] To resist recursive re-enclosure, Chapter 8 proposed targeted measures: statutory audit rights, fiduciary obligations for data stewards, and enforceable contestation rights for civil society actors.

9.3.3 Democratic Governance as Resistance

In conditions often described as 'post-truth', enclosure does not merely restrict access to data: it erodes public agency over the meaning of information. Commons-based governance provides a counter-logic by embedding plurality, accountability, and co-production into institutional design. Properly implemented, mechanisms such as open licensing carve-outs, transparency mandates, and procurement-linked openness re-assert knowledge as a public good. These reforms do not eliminate conflict but preserve the conditions for contestation—an essential safeguard for democratic legitimacy in the governance of AI.

Conclusion to Chapter 9 and the Dissertation

This dissertation asked how law and policy can resist the multi-layered enclosures of AI governance without undermining innovation. The analysis shows that wholesale abolition of intellectual property is not feasible under the current international order: TRIPS obligations are treaty-locked and enforceable through WTO dispute settlement, and domestic law embeds IP into corporate liability structures. As Baker notes, even TRIPS flexibilities such as compulsory licensing depend on political will rarely present in non-

[127] Shoshana Zuboff, *The Age of Surveillance Capitalism* (Profile Books 2019) 197–231.

[128] Julie E Cohen, *Between Truth and Power* (Oxford University Press 2019) 120–125.

producing states.[129] Sell likewise demonstrates that campaigns for reform have faced entrenched resistance.[130]

Within these constraints, partial commons reforms provide the only realistic path. Treaty-compliant measures already exist: compulsory licensing under TRIPS arts 30–31, disclosure and audit obligations under the EU AI Act (arts 16–29), interoperability and data-sharing under the Digital Markets Act (arts 6(4)–(9)), and procurement rules mandating open licensing of publicly funded AI outputs. These embed openness, auditability, and equitable access while preserving incentives for capital-intensive research.

The dissertation's contribution is therefore not to advocate abolition but to demonstrate how partial commons tools, grounded in law and institutionally practicable, can defend shared epistemic agency. Sustaining democratic legitimacy in the age of AI requires precisely this balance: treaty-compatible reforms that constrain enclosure while preserving innovation capacity.

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[129] Baker 2004, 12–14.

[130] Sell 2002, 490–495.

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Version History

Version	Description of Changes	Epistemic Impact	Date
1	Adapted from Kahl's dissertation titled *The Third Enclosure Movement: Rethinking AI Regulation through a Taxonomy of Restricted Knowledge in the UK, EU, US—and China*, and released under revised title *The Third Enclosure Movement: Rethinking AI Regulation through a Taxonomy of Restricted Knowledge*	N/A	2025-11-06

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